

CSCI 598H

Applied Computer Vision for Engineering Systems

Lecture 1-1: Introduction

Dr. Minghan Li

Assistant Professor

Computer Science Department

Colorado School of Mines

Aug/25/2026



About Me

Assistant Professor @ CS

Colorado School of Mines

Email: minghan.li@gmail.com

Office: CTLM 214A

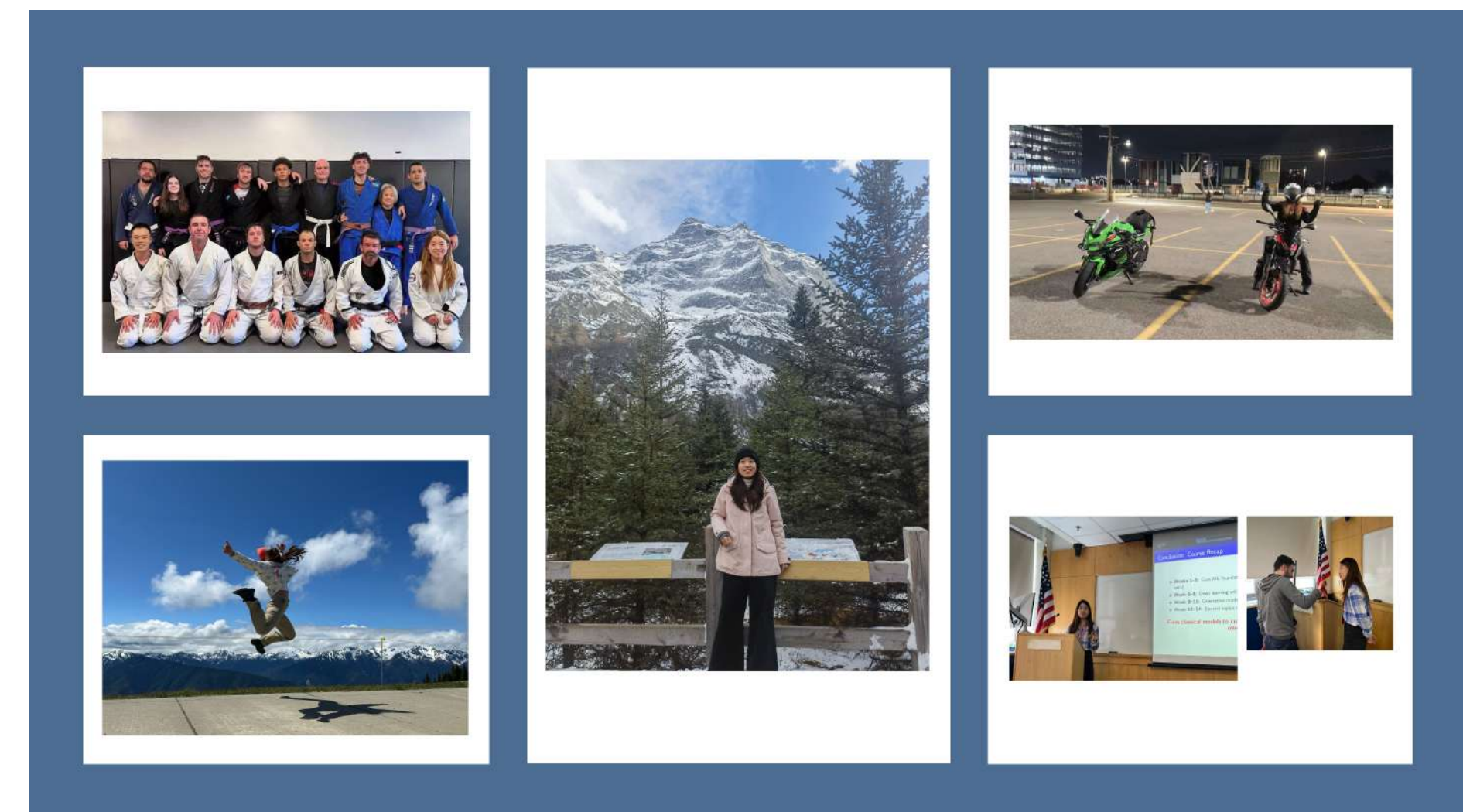
Office hours: Tue/Thur 2-4pm

Dr. Minghan Li (or Niki)

- Postdoc in Harvard AI & Robotic Lab @ Harvard Medical School
- PhD in Computer Science @ Hong Kong PolyU
- Master in Statistics @ Xi'an Jiaotong University
- Bachelor in Mathematics @ Xinjing University



Harvard AI & Robotic Seminar



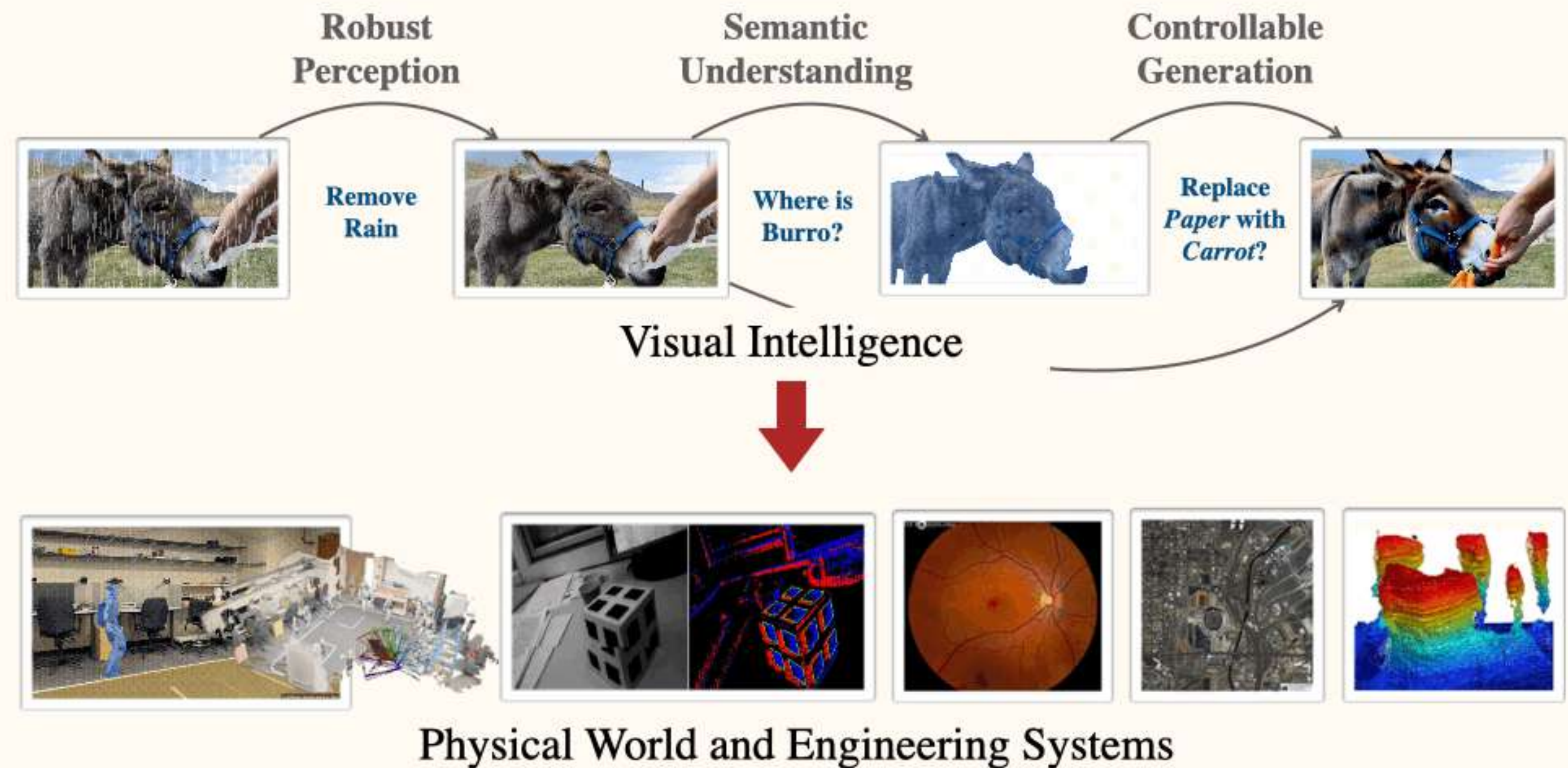
BJJ / Hiking / Motorcycle / Teaching

MAILab: Computer Vision & AI

Research Interests

- **Computer Vision Tasks**
 - Visual understanding / generation / editing
 - Foundation models
- **Physical AI**
 - Robotic perception
 - Embodied reasoning
- **AI for Engineering**
 - Mining & Energy
 - Remote sensing
 - Material science

Visual Intelligence For Physical and Engineered World



Quick Class Survey

Help me get to know you !

Your background · Research interests · What you want to learn

[https://www.mentimeter.com/app/presentation/
alh1kc6cnoe4vbi5u7j9gertq5wt67bu/edit?
question=xkmgstzadg9t](https://www.mentimeter.com/app/presentation/alh1kc6cnoe4vbi5u7j9gertq5wt67bu/edit?question=xkmgstzadg9t)

Welcome to CSCI 598H

Applied Computer Vision for Engineering Systems

Welcome to CSCI 598H

Applied Computer Vision for Engineering Systems

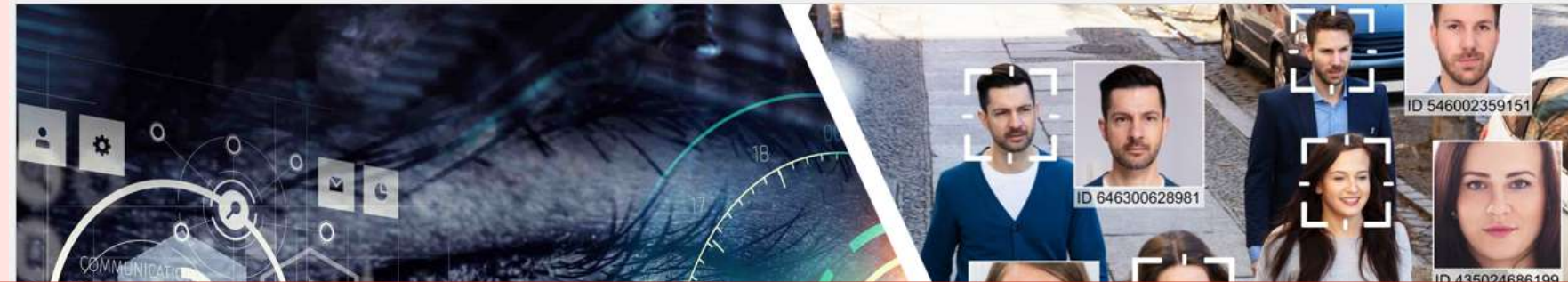
[Link](#)



Welcome to CSCI 598H

Applied Computer Vision for Engineering Systems

[Link](#)



CS231n @ Stanford

**Deep Learning for
Computer Vision**

Feifei Li & Ehsan



Welcome to CSCI 598H

Applied Computer Vision for Engineering Systems

[Link](#)

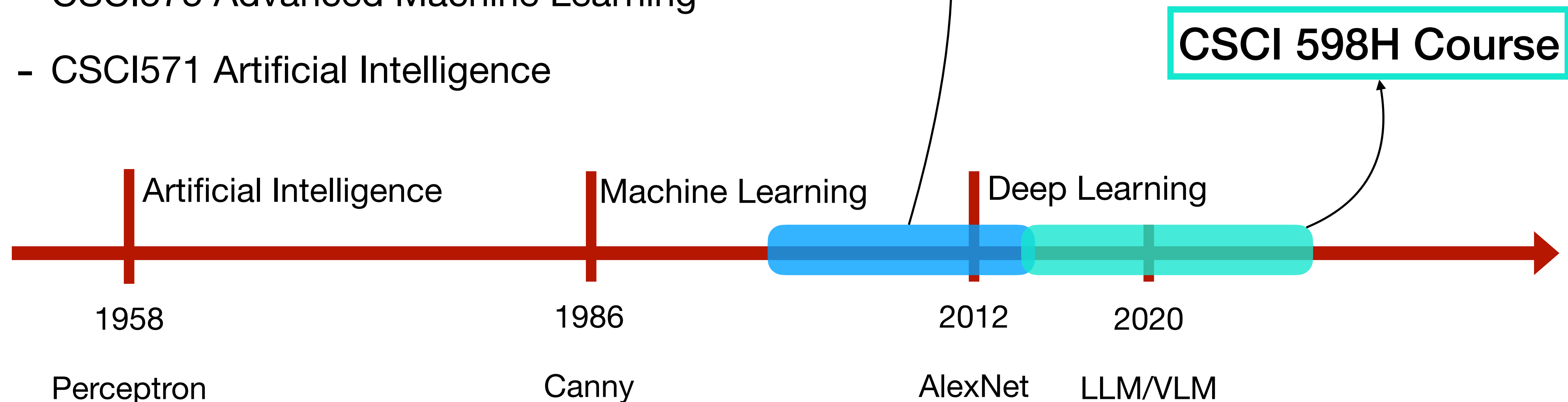


[Link](#)

Welcome to CSCI 598H

Prerequisites: completion of at least one of the following courses

- CSCI507/EENG507 Introduction to Computer Vision
- CSCI570 Introduction to Machine Learning
- CSCI576/476 Deep Learning
- CSCI470 Introduction to Machine Learning
- CSCI575 Advanced Machine Learning
- CSCI571 Artificial Intelligence



CSCI 598H Course Overview

Weeks 2-4

Weeks 5-11

Weeks 12-16

Module 1

Deep Learning Basics

- Linear Classifier and Optimization
- CNNs and BackPropagation
- RNNs, Attention, Transformers

Module 2

Modern Visual Intelligence

- Perceiving and understanding the visual world
- Generative models and foundation models
- Geometry, 3D vision and spatial understanding

Module 3

Engineering Systems

- Robotics
- Earth science
- Medical imaging
- Energy

Today's agenda

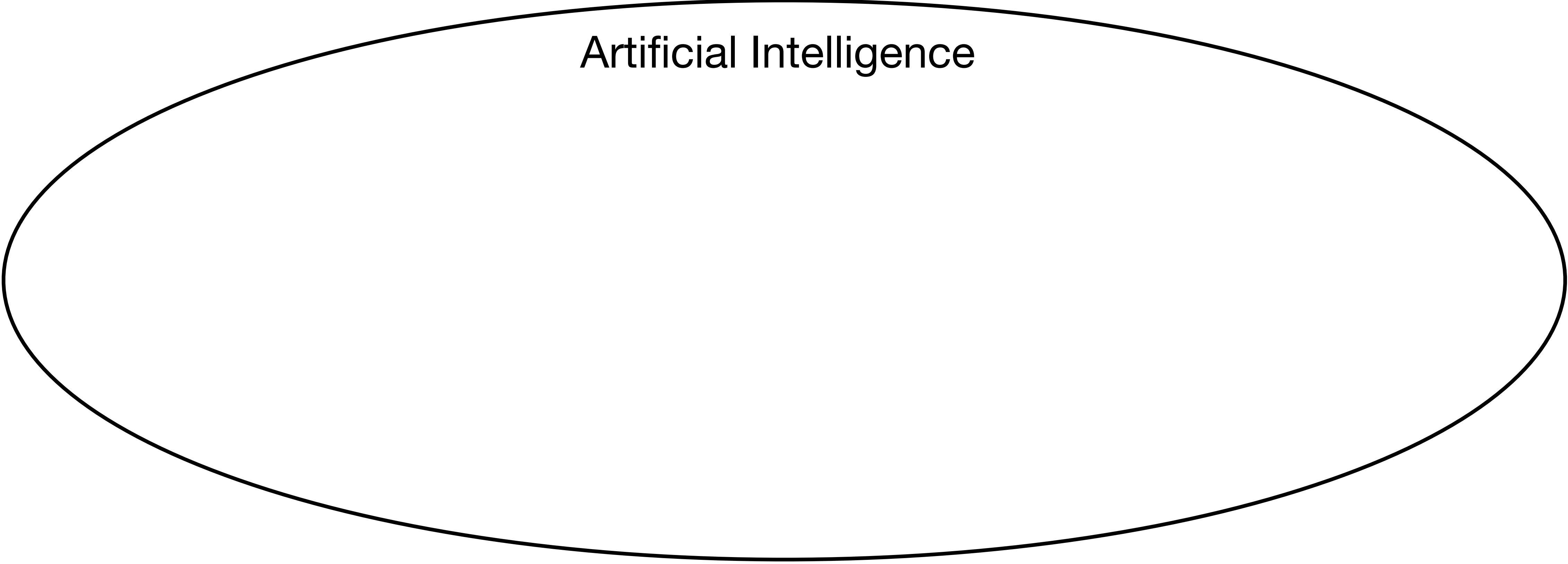
- A brief history of computer vision (CV)
- Examples of CV for engineering applications
- CSCI 598H course overview

Today's agenda

- A brief history of computer vision
- Examples of CV for engineering applications
- CSCI 598H course overview

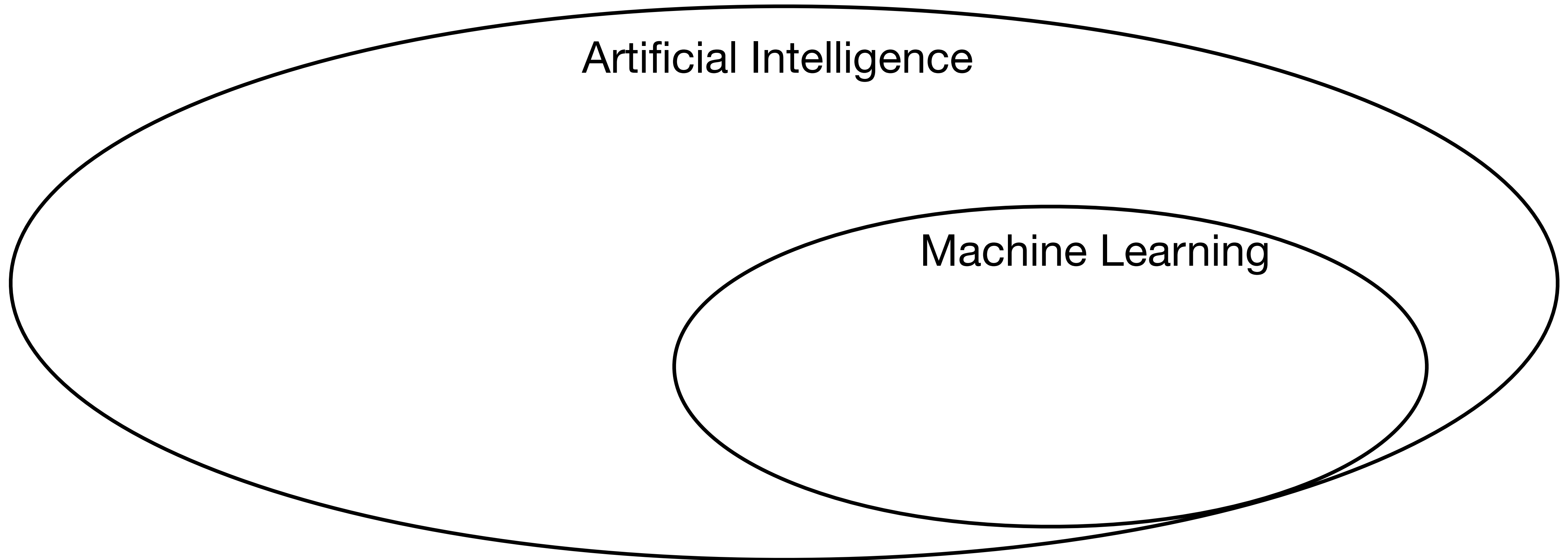
Course Scope

Artificial Intelligence



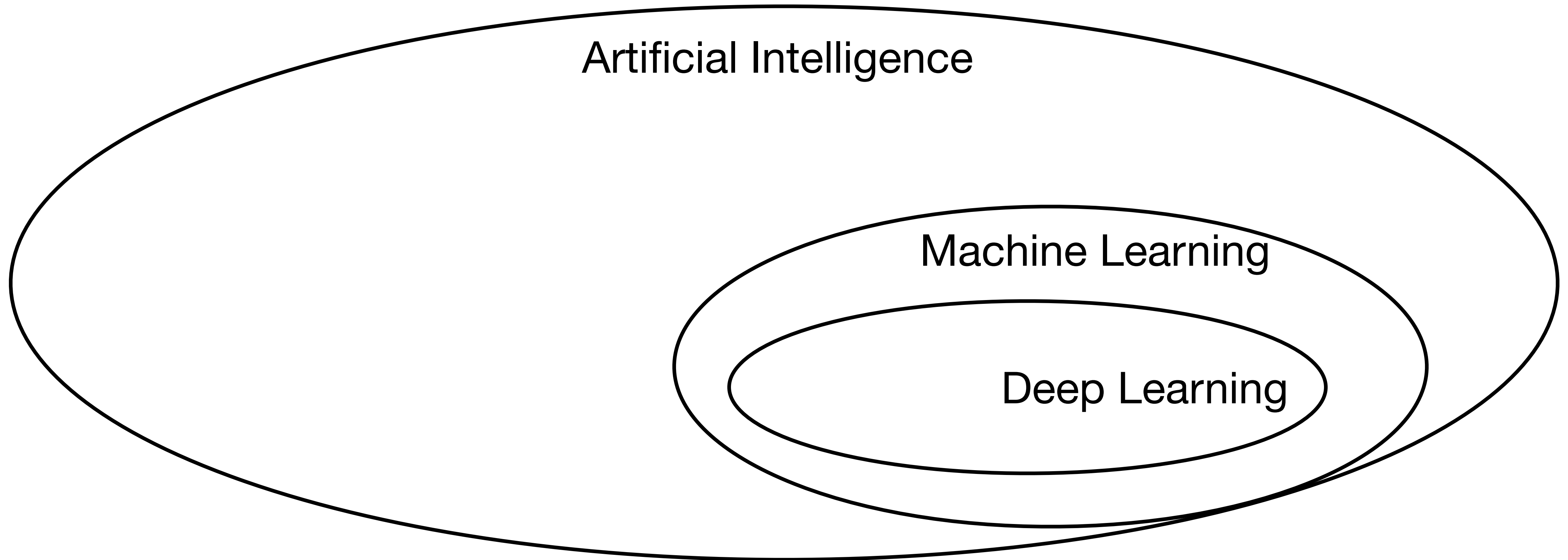
AI: building machines or computational systems that can perform tasks that **normally require some form of human intelligence**

Course Scope



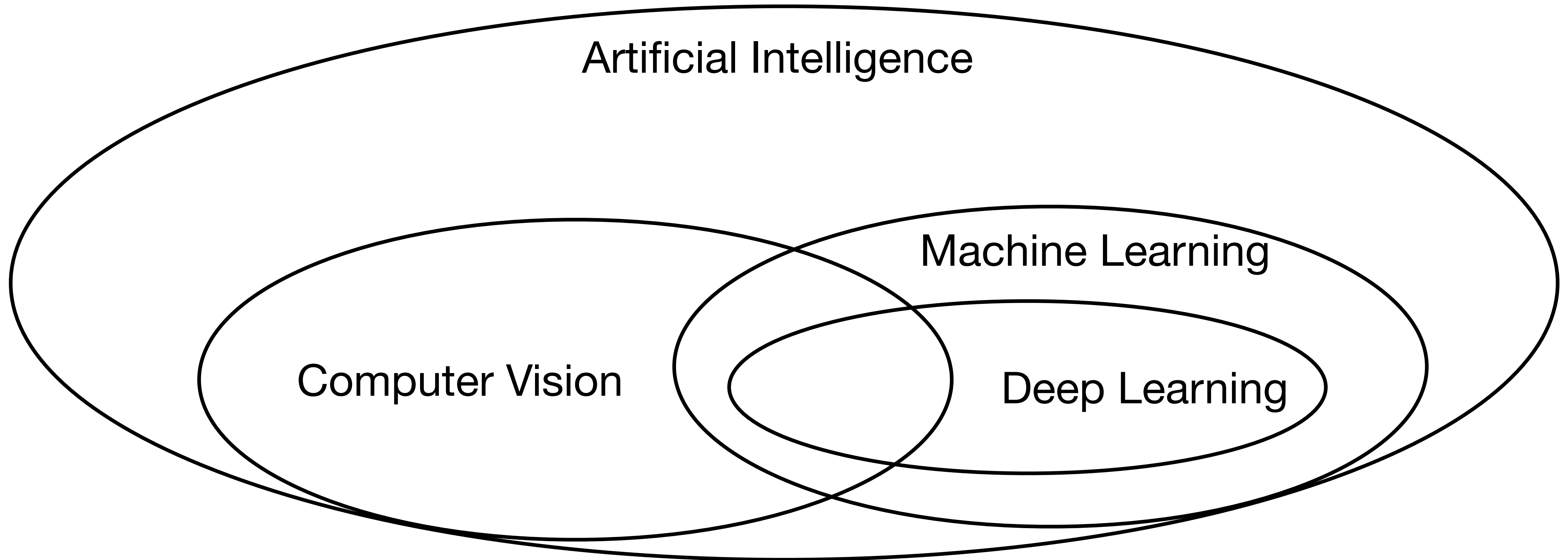
ML: instead of manually programming all the rules, we allow machines to **learn patterns and functions from data**

Course Scope



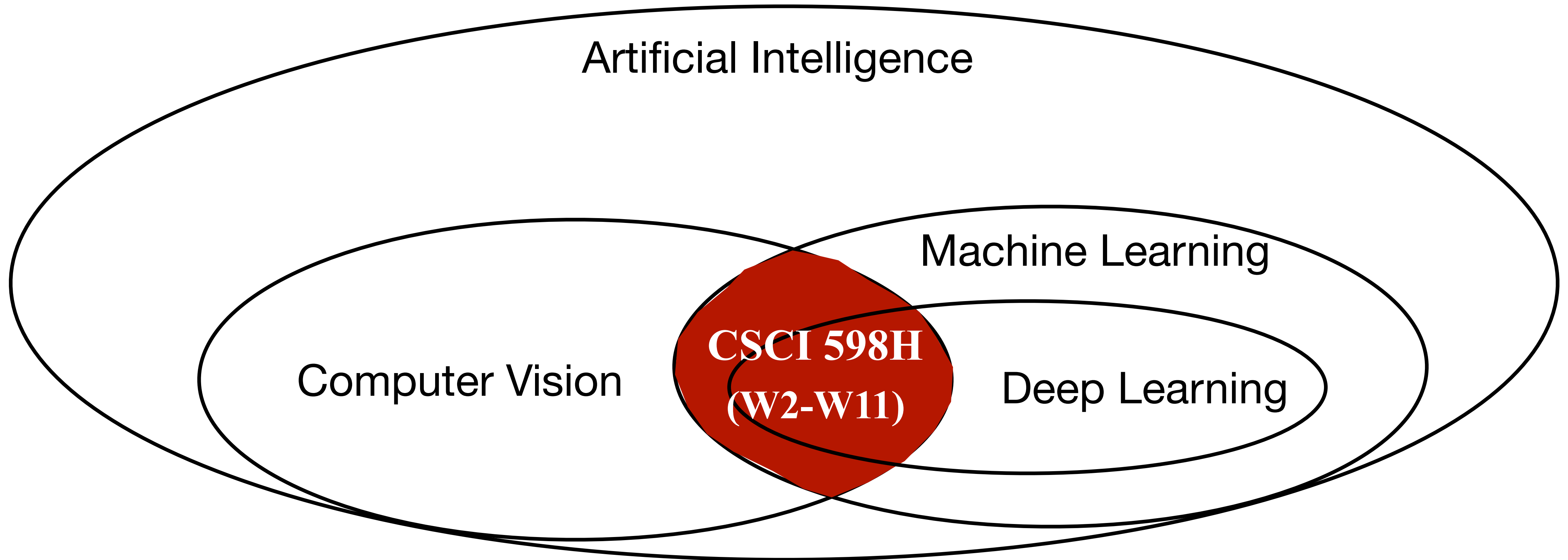
DL: use multi-layer neural networks to learn increasingly complex representations from data

Course Scope



CV: perceive, understand, and reason about visual information, such as images and videos.

Course Scope



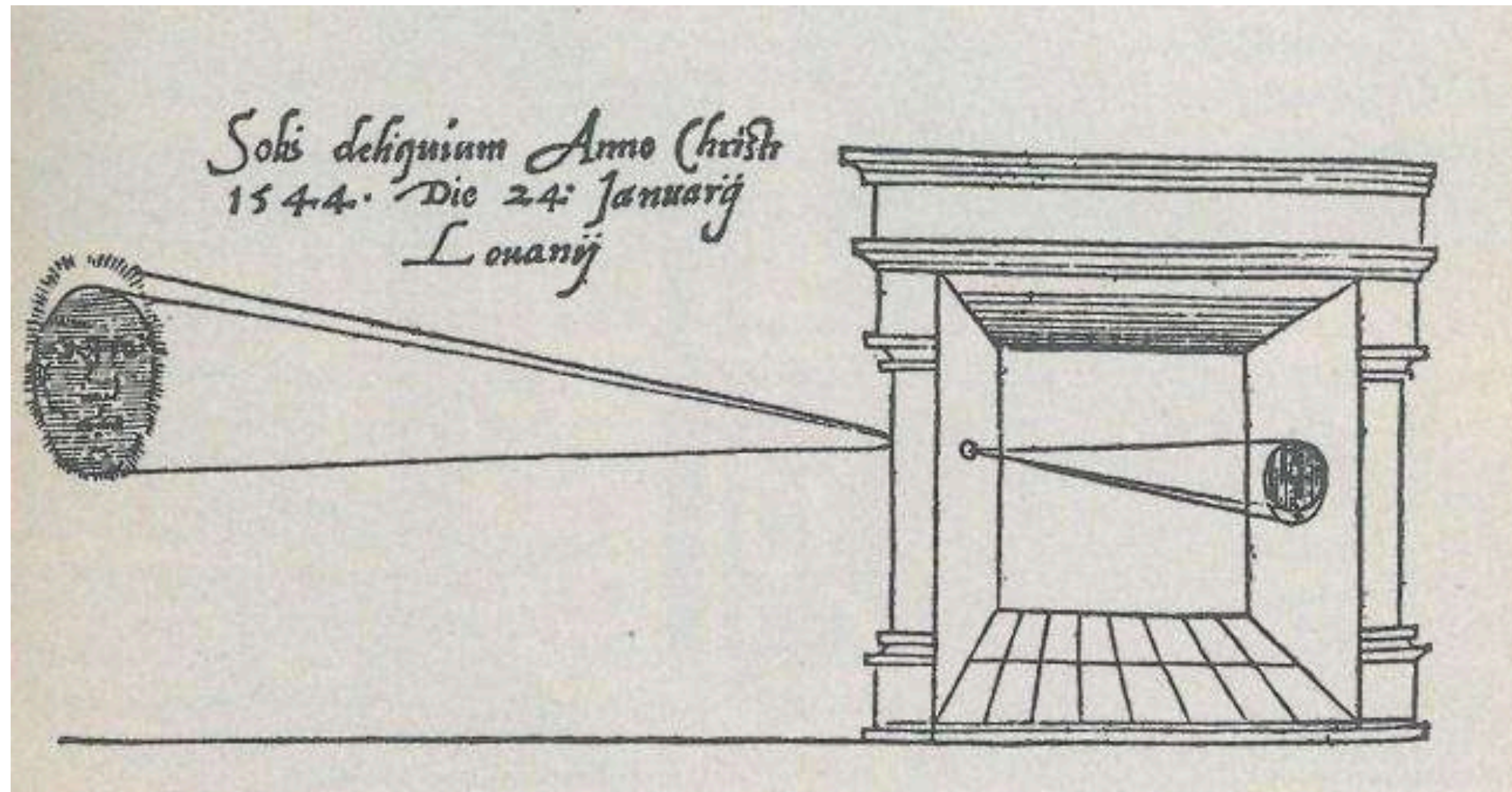
Vision



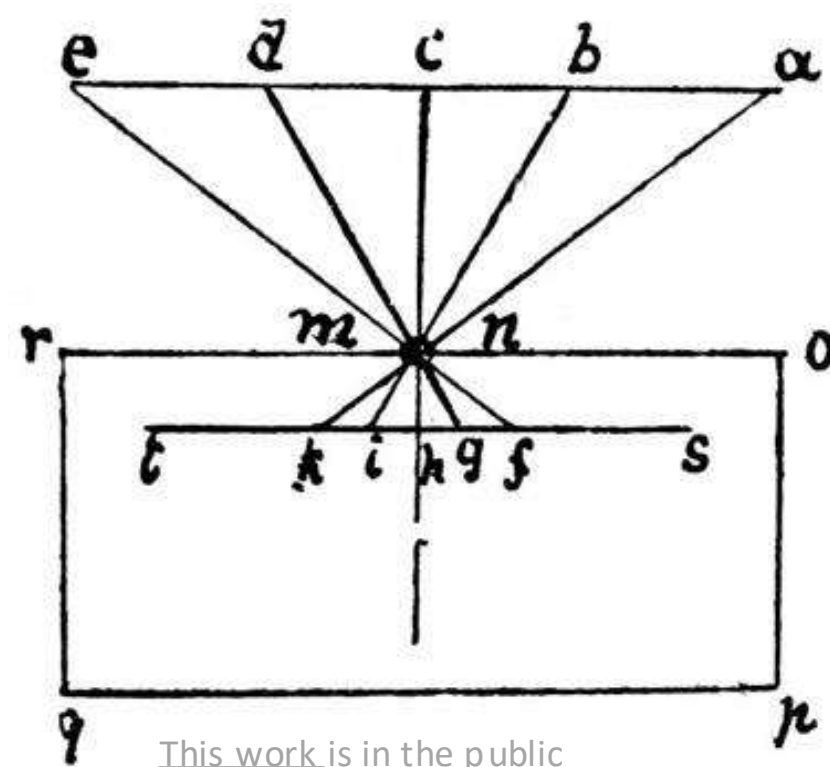
Fundamental purpose: to acquire information about the surrounding world.

Camera Obscura

Gemma Frisius, 1545



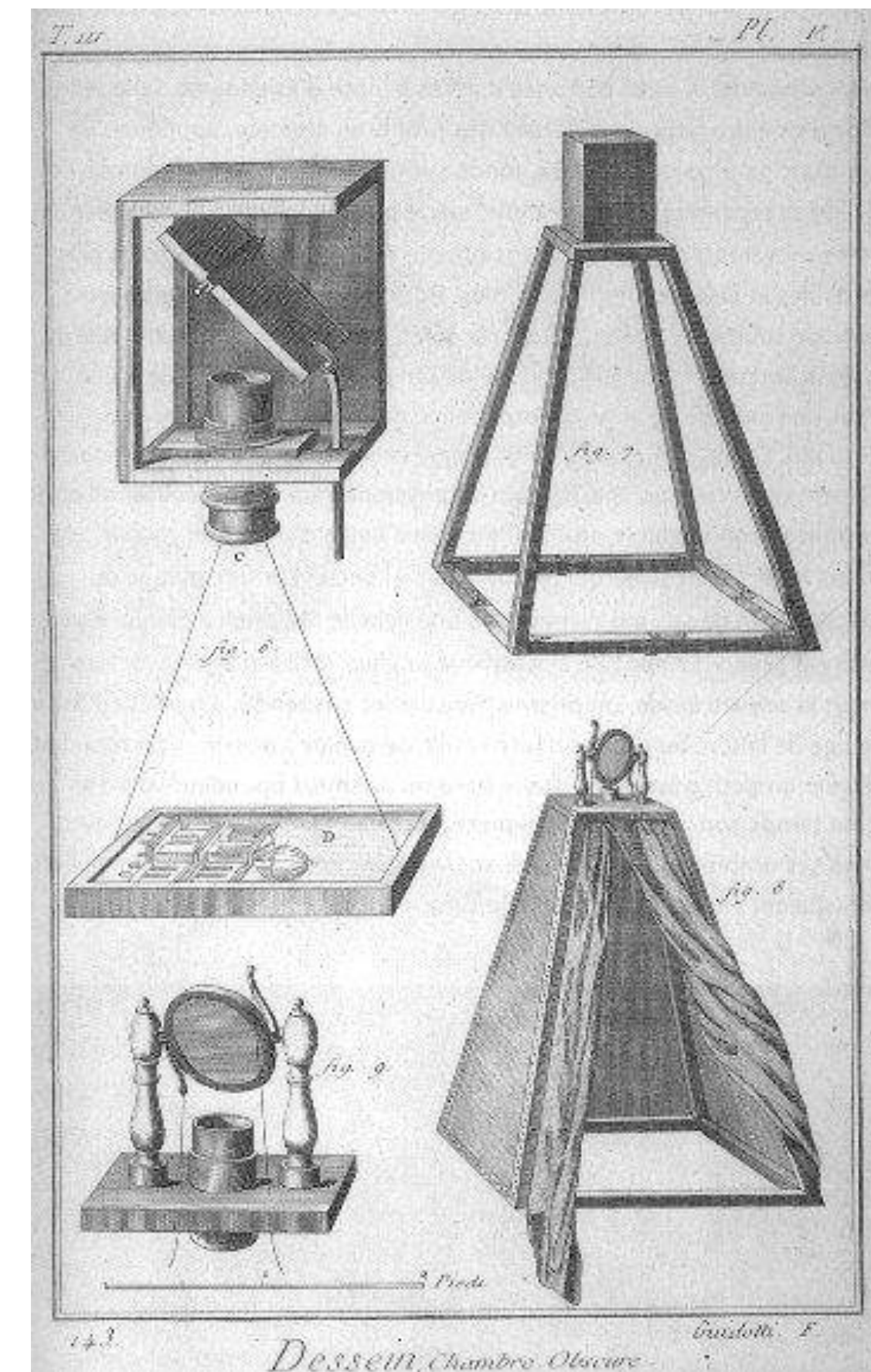
This work is in the public domain



This work is in the public domain

Leonardo da Vinci,
16th Century AD

Encyclopedia, 18th Century



This work is in the public domain

Light from the outside scene passes through the small hole and forms an inverted image on the opposite surface.

Human Vision vs Computer Vision



What we see

Human Vision vs Computer Vision



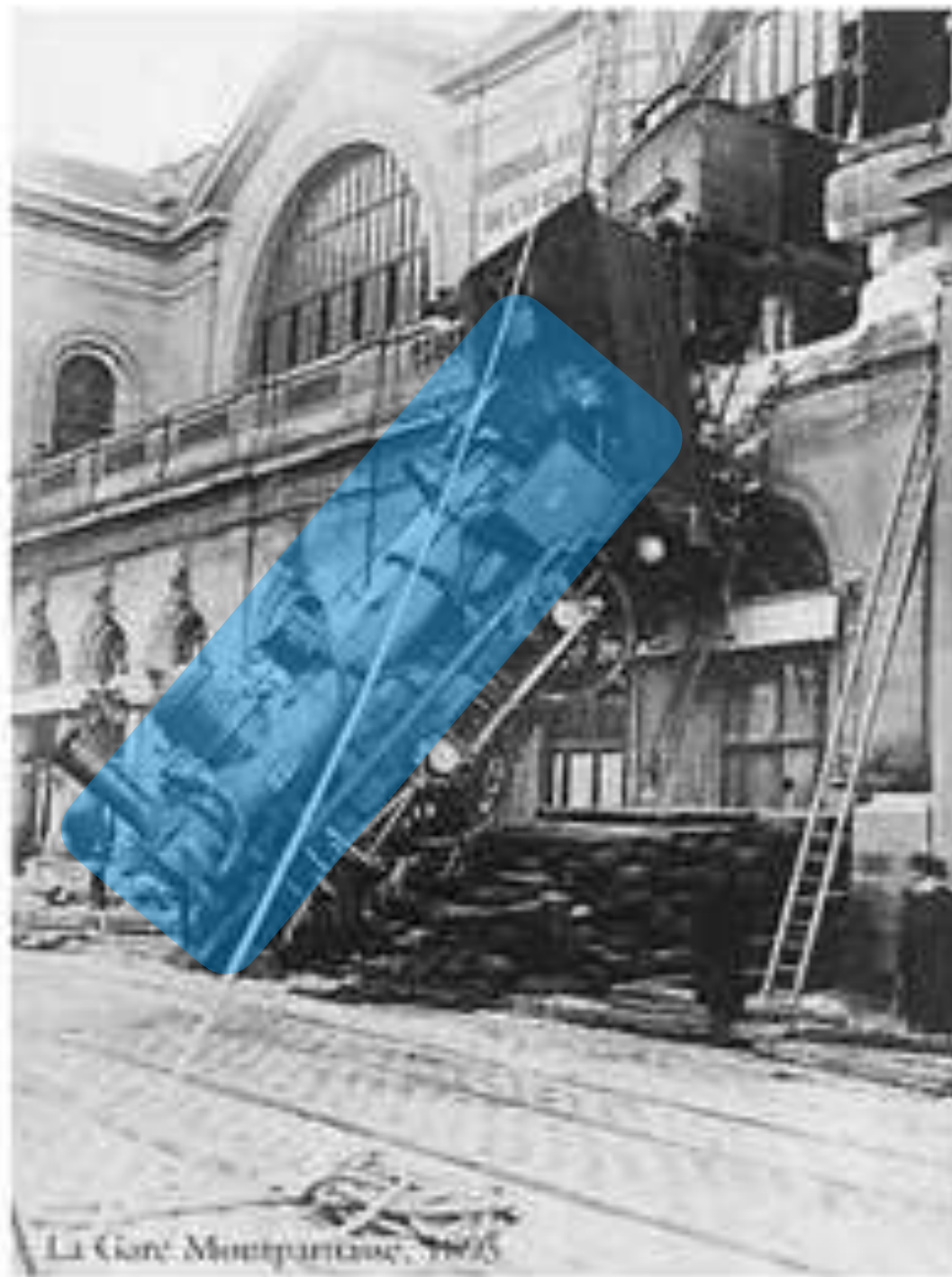
What we see

numbers / numerical pixel values

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What a computer sees

Human Vision vs Computer Vision



What we see

numbers / numerical pixel values

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0			3	4	5
4	3	2			3	2	5	4
7	4			3	0	1	2	3
6	5			2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

How do we go from low-level numerical values to high-level semantic understanding?

Computer Vision is Everywhere!



More examples please



Computer Vision is everywhere!



Left to right:
[Image by Roger H Goun](#) is licensed under [CC BY 2.0](#)
[Image](#) is [CC0 1.0](#) public domain
[Image](#) is [CC0 1.0](#) public domain
[Image](#) is [CC0 1.0](#) public domain



Left to right:
[Image](#) is free to use
[Image](#) is [CC0 1.0](#) public domain
[Image](#) by [NASA](#) is licensed under [CC BY 2.0](#)
[Image](#) is [CC0 1.0](#) public domain



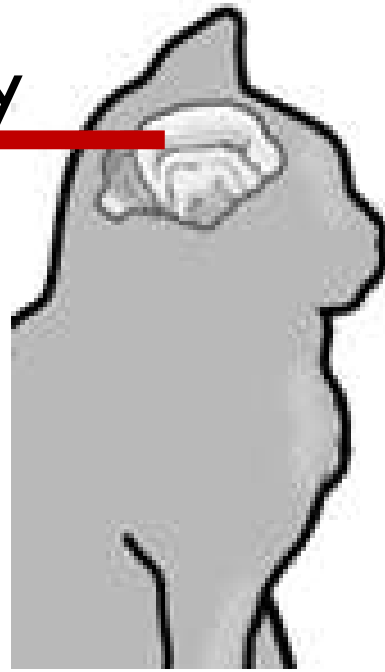
Bottom row, left to right
[Image](#) is [CC0 1.0](#) public domain
[Image](#) by [Derek Keats](#) is licensed under [CC BY 2.0](#); changes made
[Image](#) is public domain
[Image](#) is licensed under [CC-BY 2.0](#); changes made

Computer vision today looks extremely powerful.

Where did we come from?

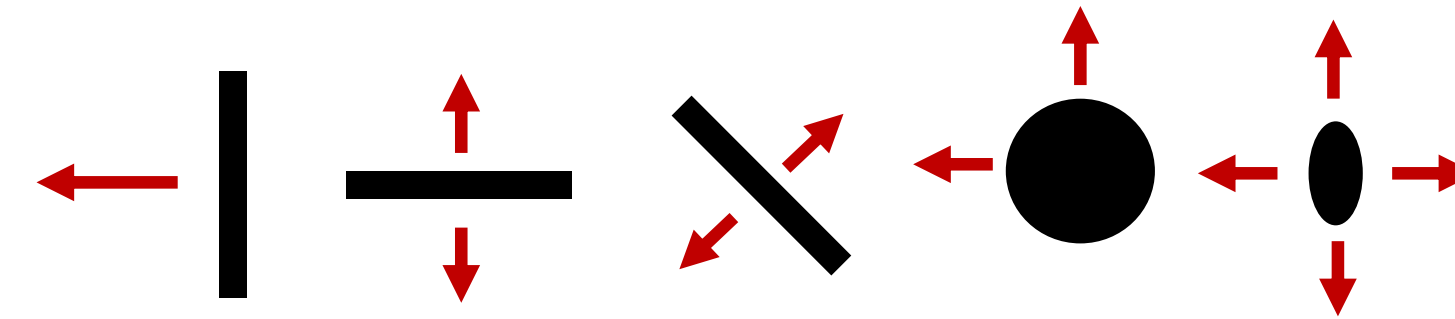
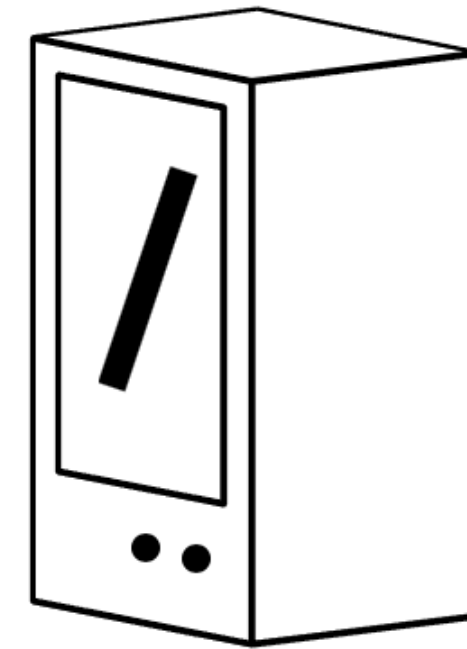
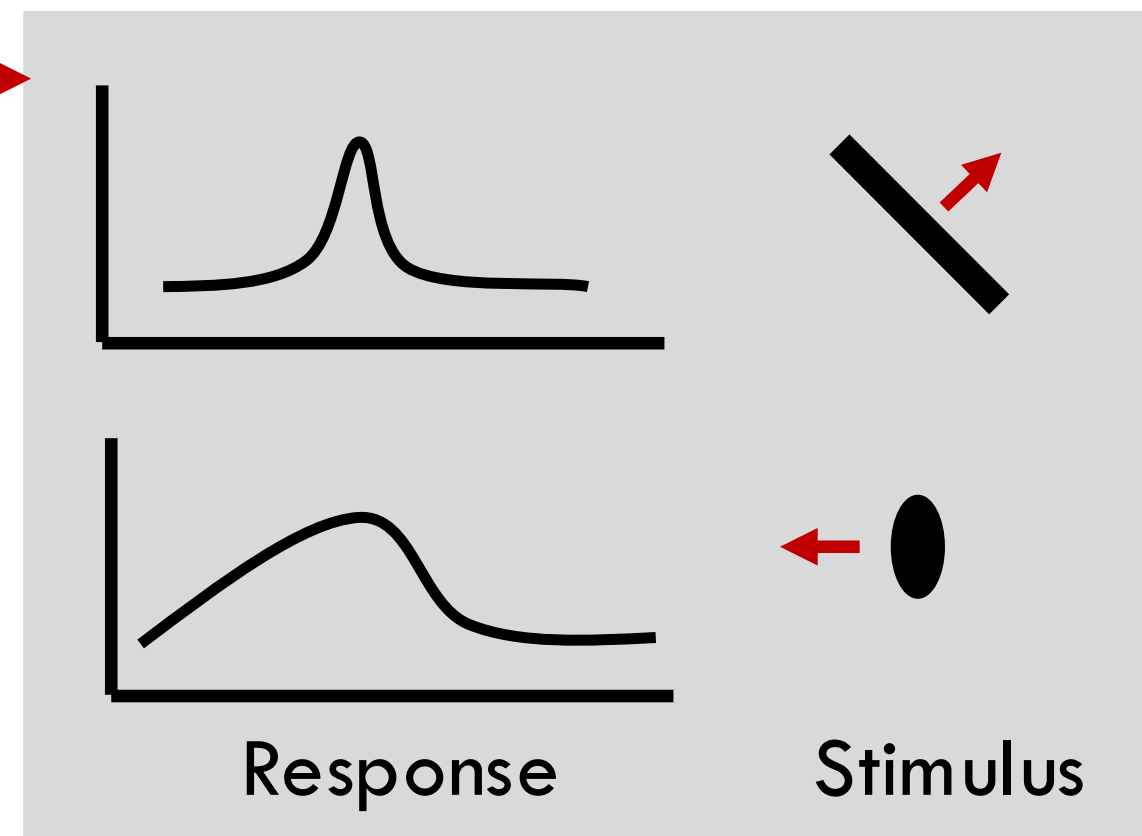
Hubel and Wiesel, 1959

Measure
brain activity



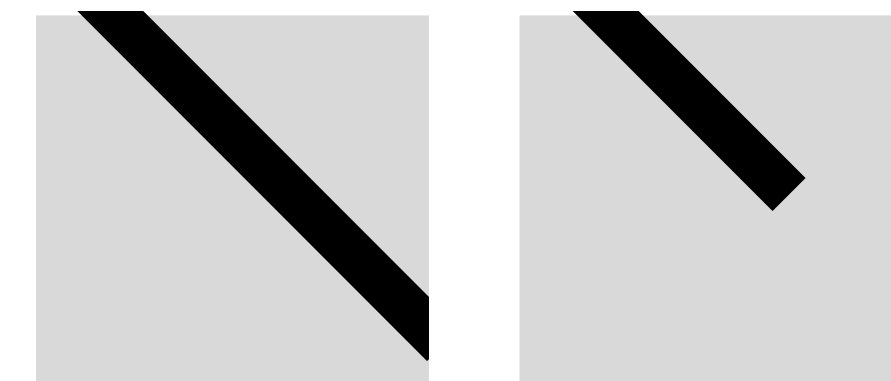
Cat image by CNX OpenStax is licensed under CC BY 4.0; changes made

1959
Hubel & Wiesel



Simple cells:
Response to specific
rotation and orientation

Complex cells:
Response to light
orientation and
movement, some
translation invariance



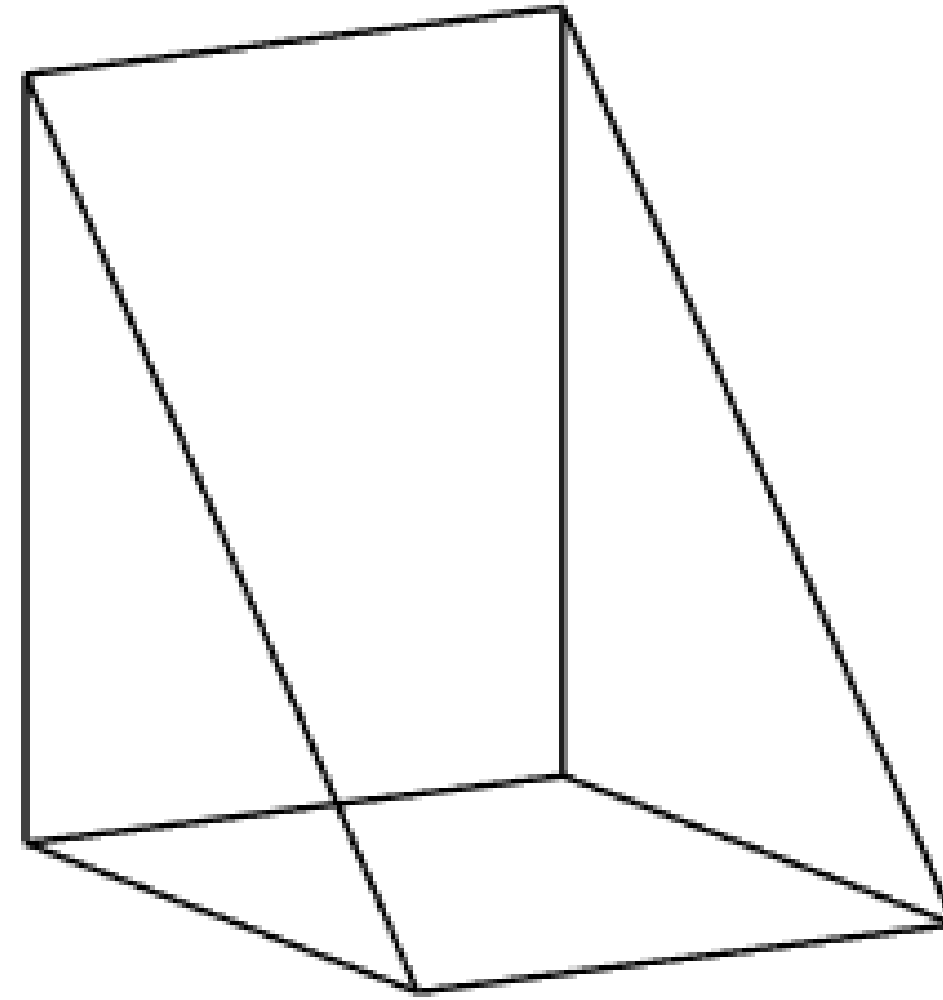
No response

Neuron Science: visual processing may be hierarchical, from simple patterns to more complex representations.

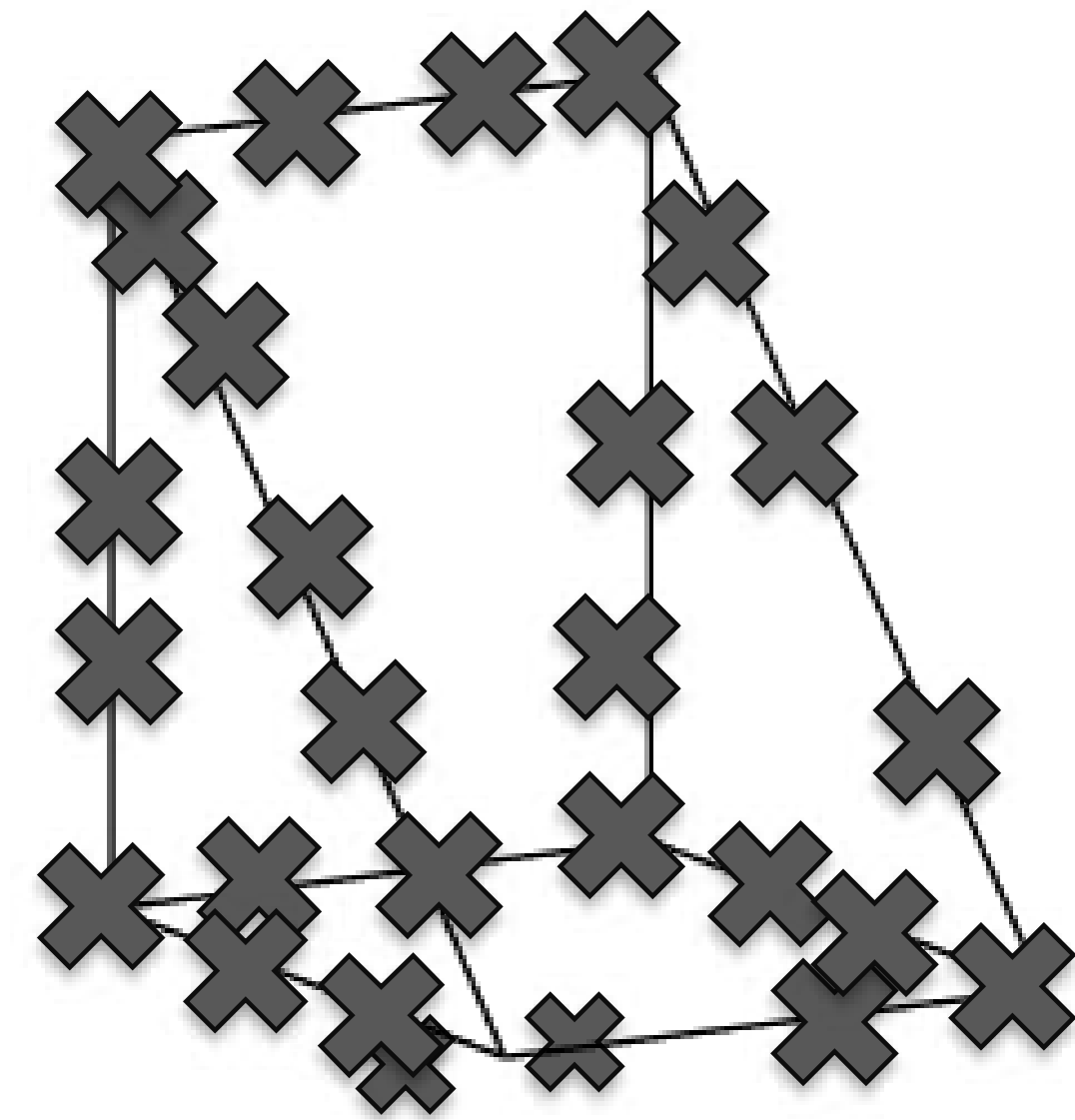
Larry Roberts, 1963



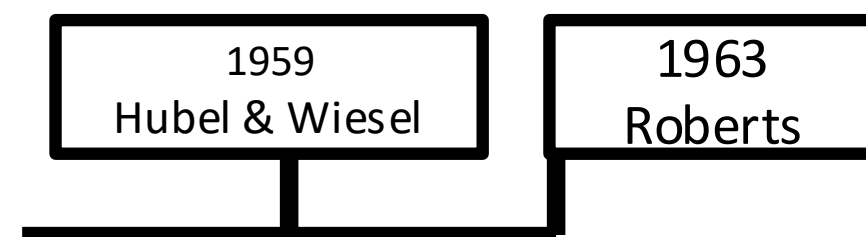
(a) Original picture



(b) Differentiated picture



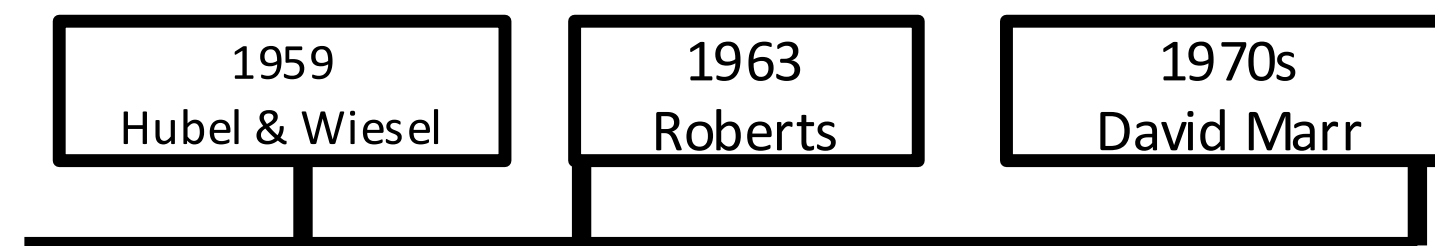
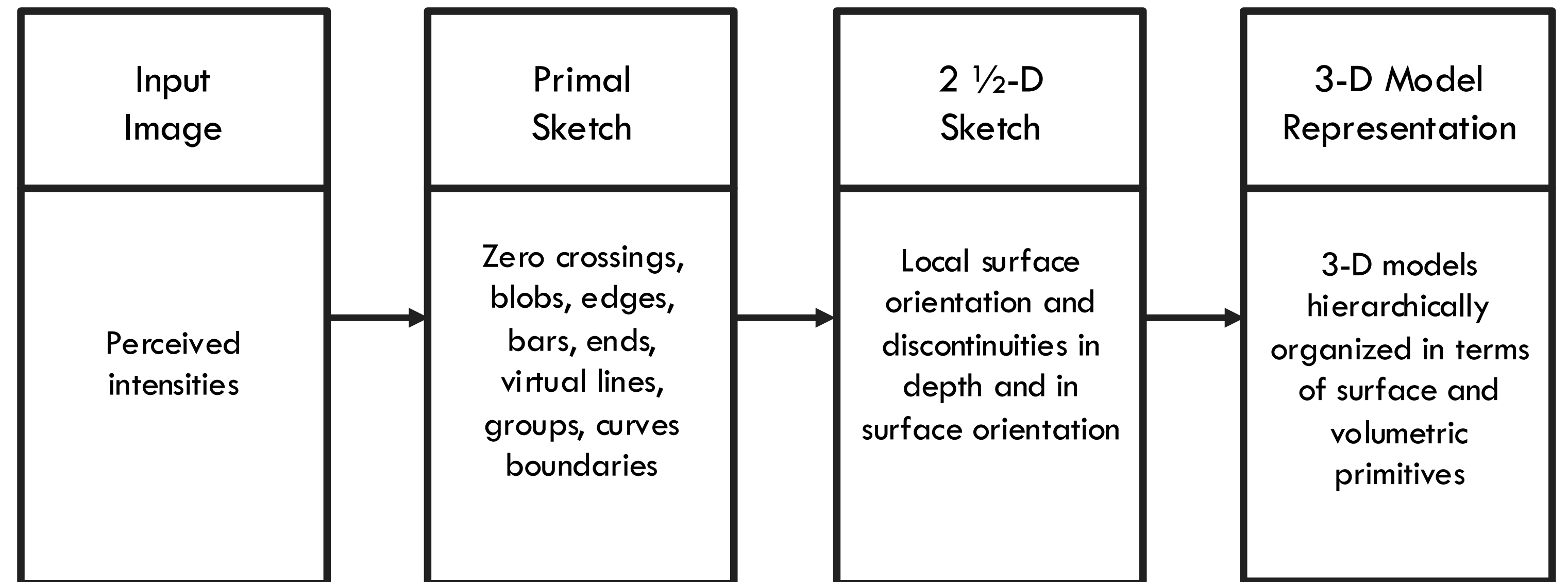
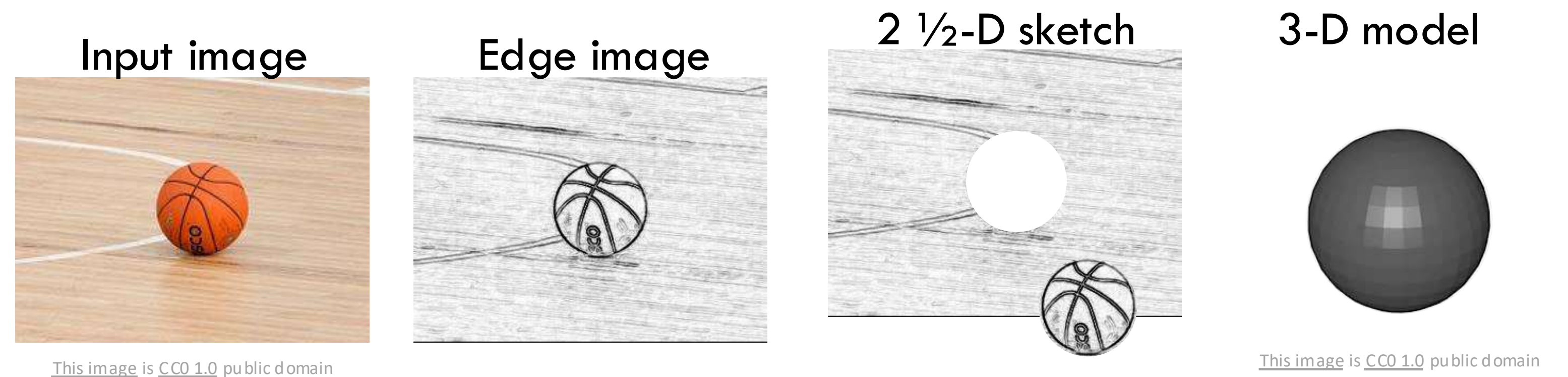
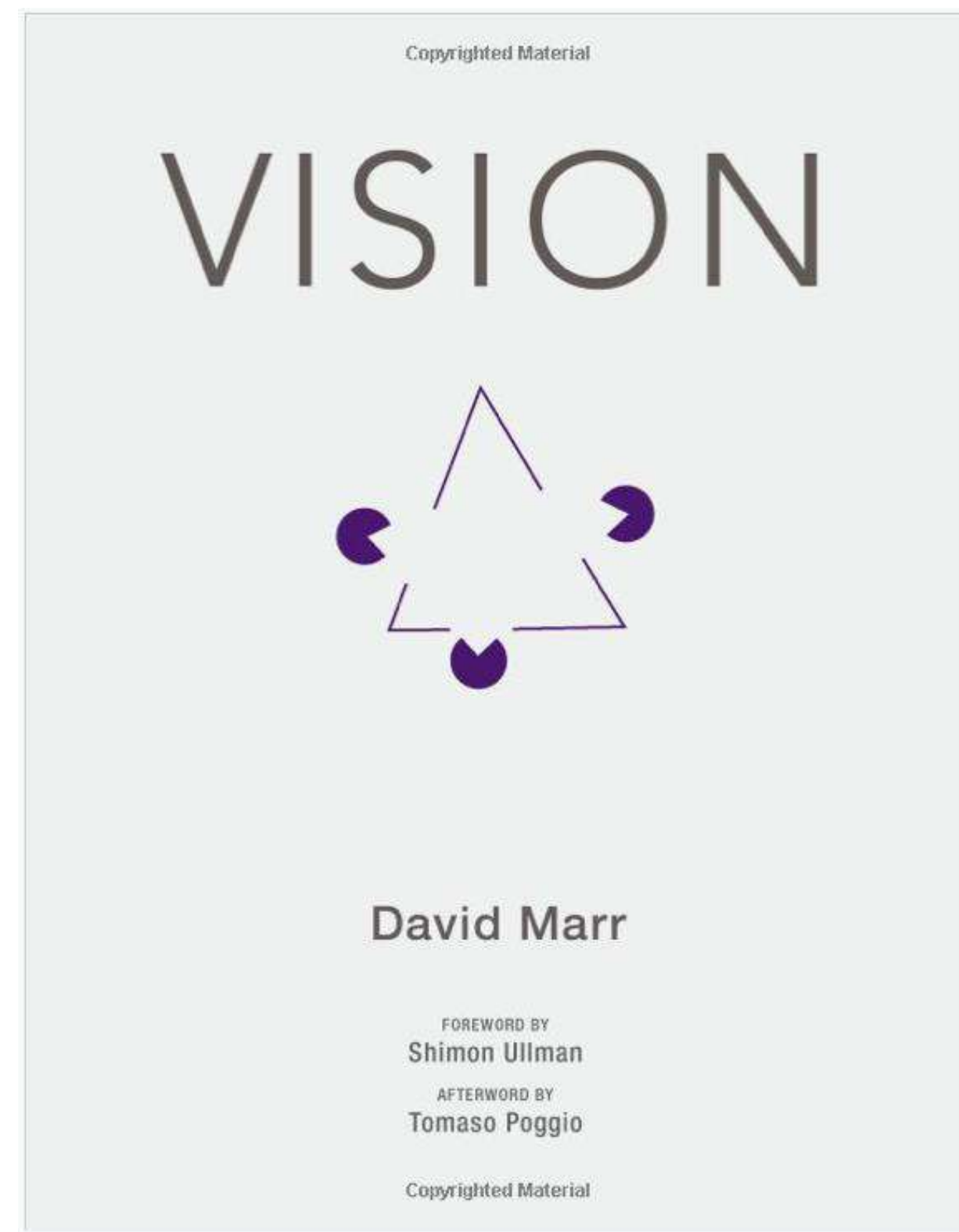
(c) Feature points selected



Lawrence Gilman Roberts, "Machine Perception of Three-Dimensional Solids", 1963

Slide inspiration: Justin Johnson

To recover 3D structure from images: simple edges and feature points could be used to reason about the shape of an object.

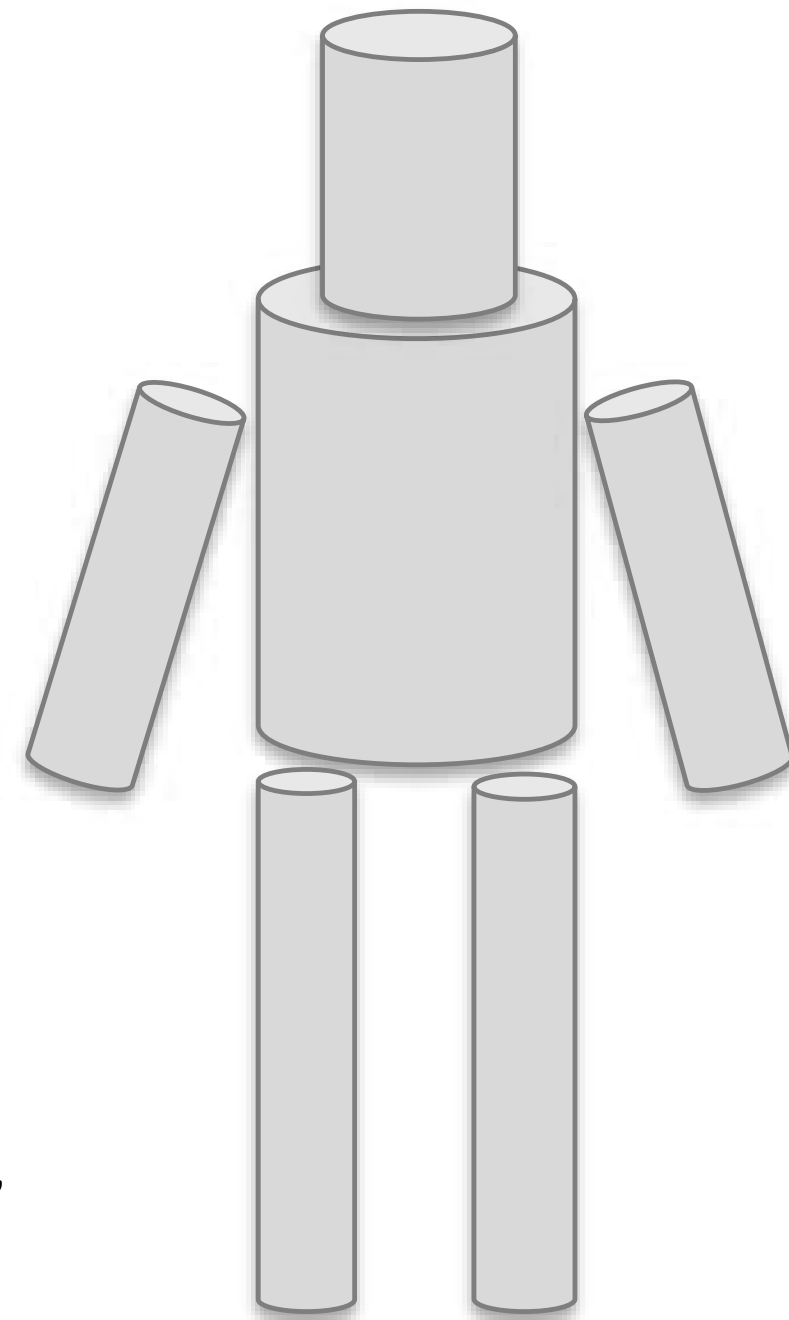


Stages of Visual Representation, David Marr, 1970s

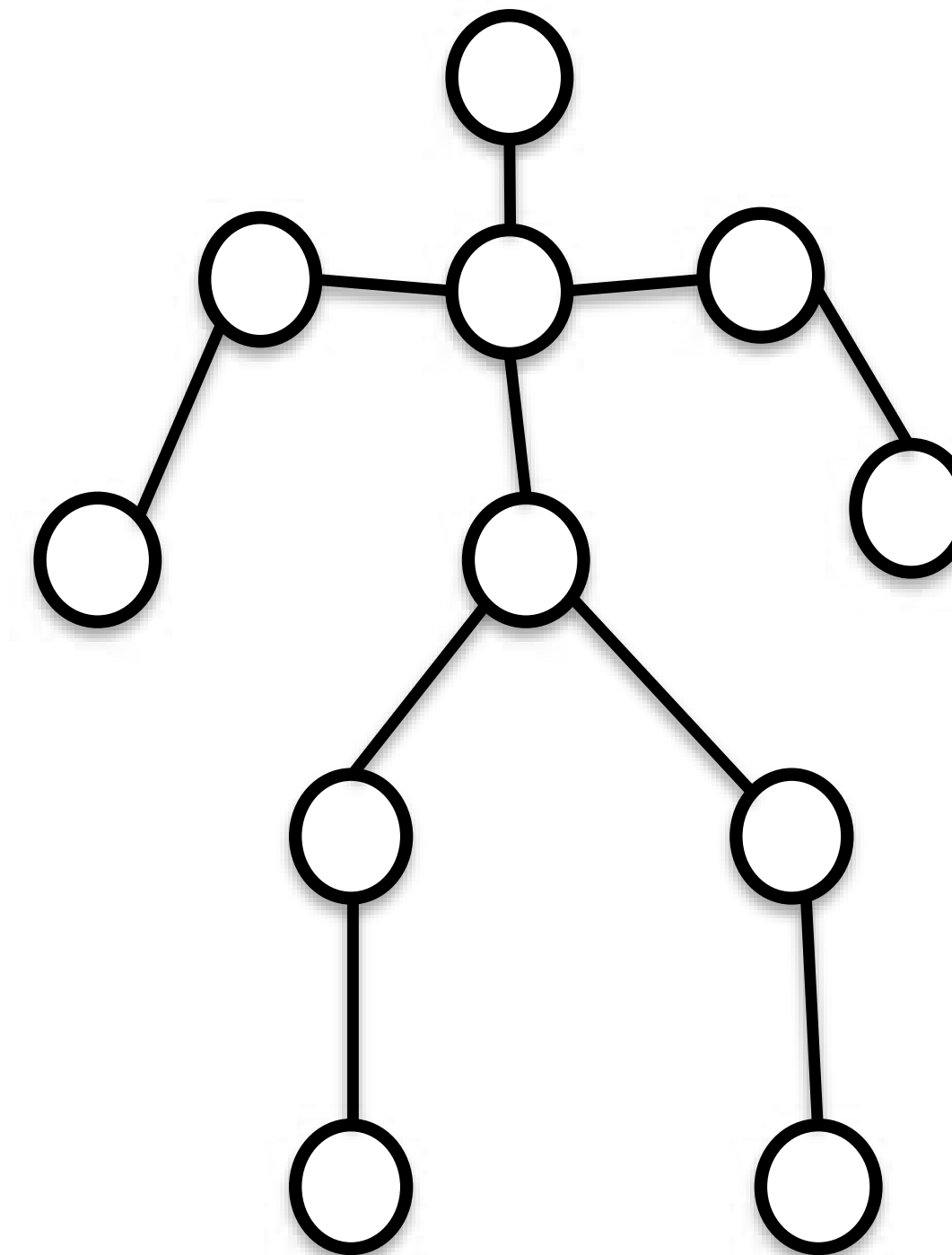
Slide inspiration: Justin Johnson

Vision happens through multiple stages of representation.
 Numbers —> Edges/surfaces —> 3D understanding of the scene

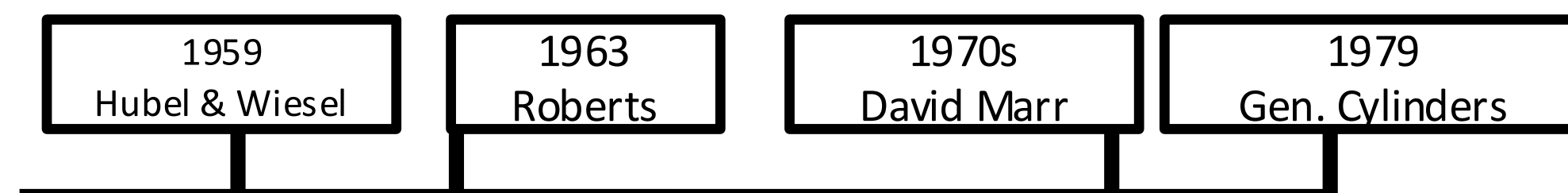
Recognition via Parts (1970s)



Generalized Cylinders,
Brooks and Binford,
1979



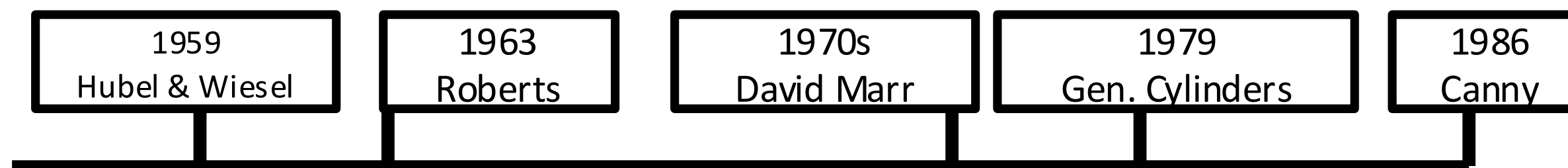
Pictorial Structures,
Fischler and Elshlager, 1973



Slide inspiration: Justin Johnson

Representation: use simpler components and their relationships

Recognition via Edge Detection (1980s)

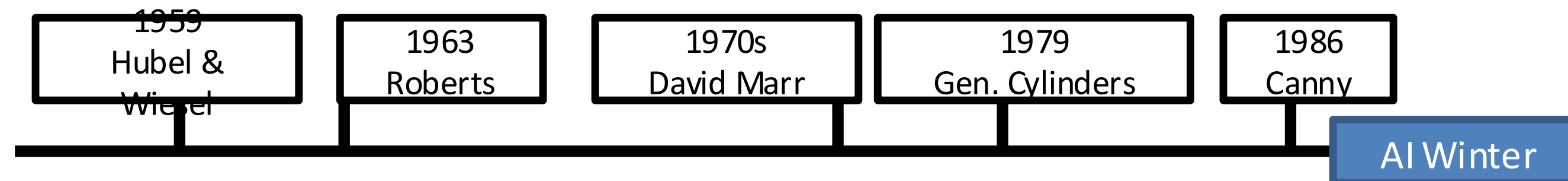


John Canny, 1986
David Lowe, 1987

Large intensity changes often correspond to object boundaries.

Arriving at an “AI winter”

- Enthusiasm (and funding!) for AI research dwindled
- “Expert Systems” failed to deliver on their promises
- But subfields of AI continues to grow
 - Computer vision, NLP, robotics, compbio, etc.

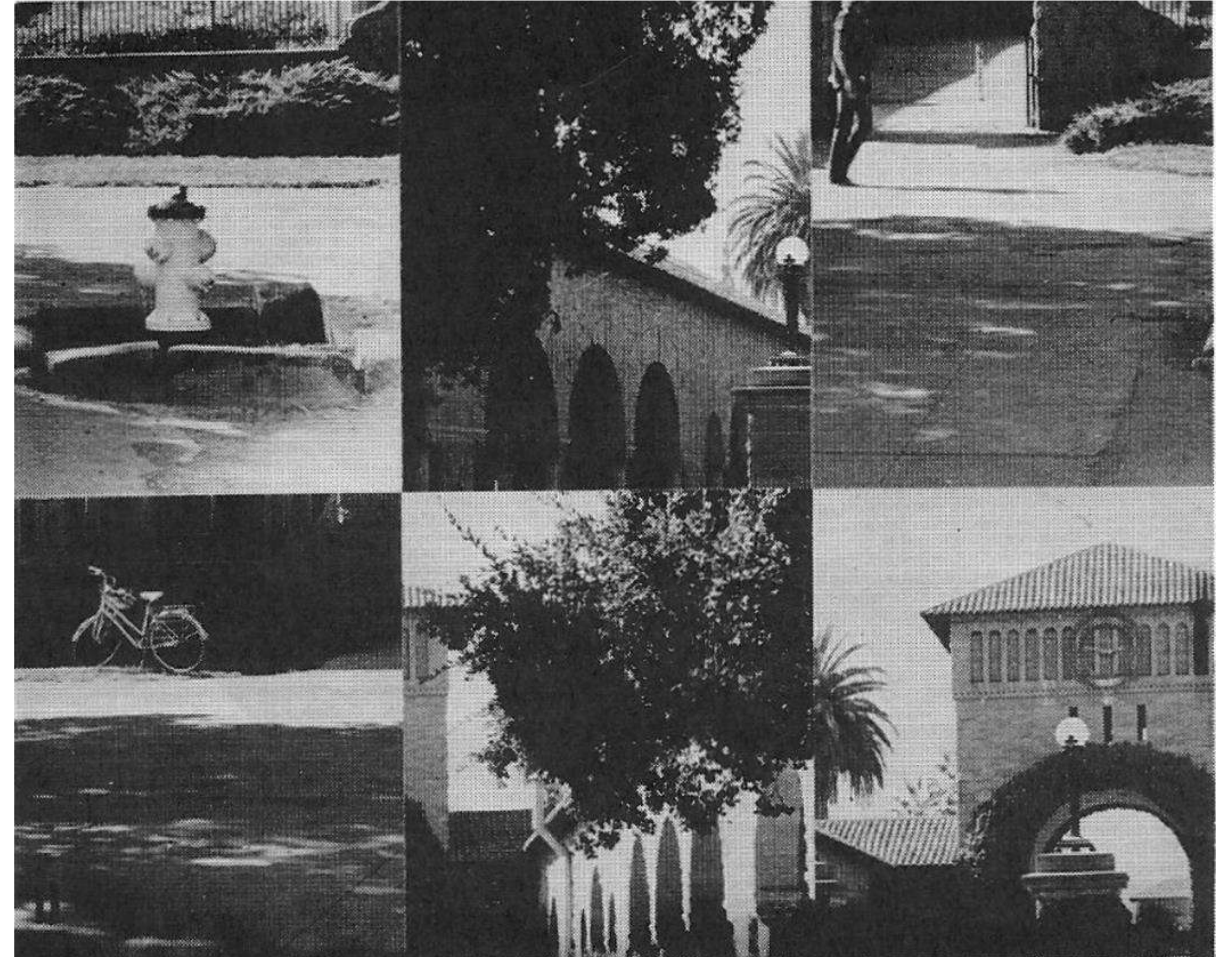


Algorithms and computing resources were limited

In the meantime...seminal work in
cognitive and neuroscience

Perceiving Real-World Scenes

Irving Biederman

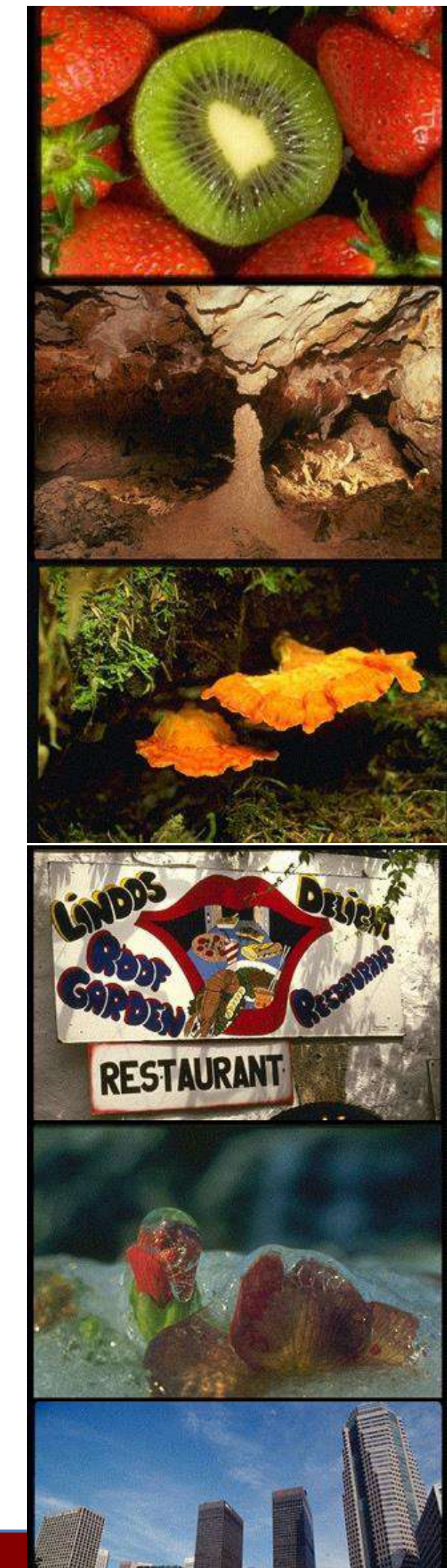
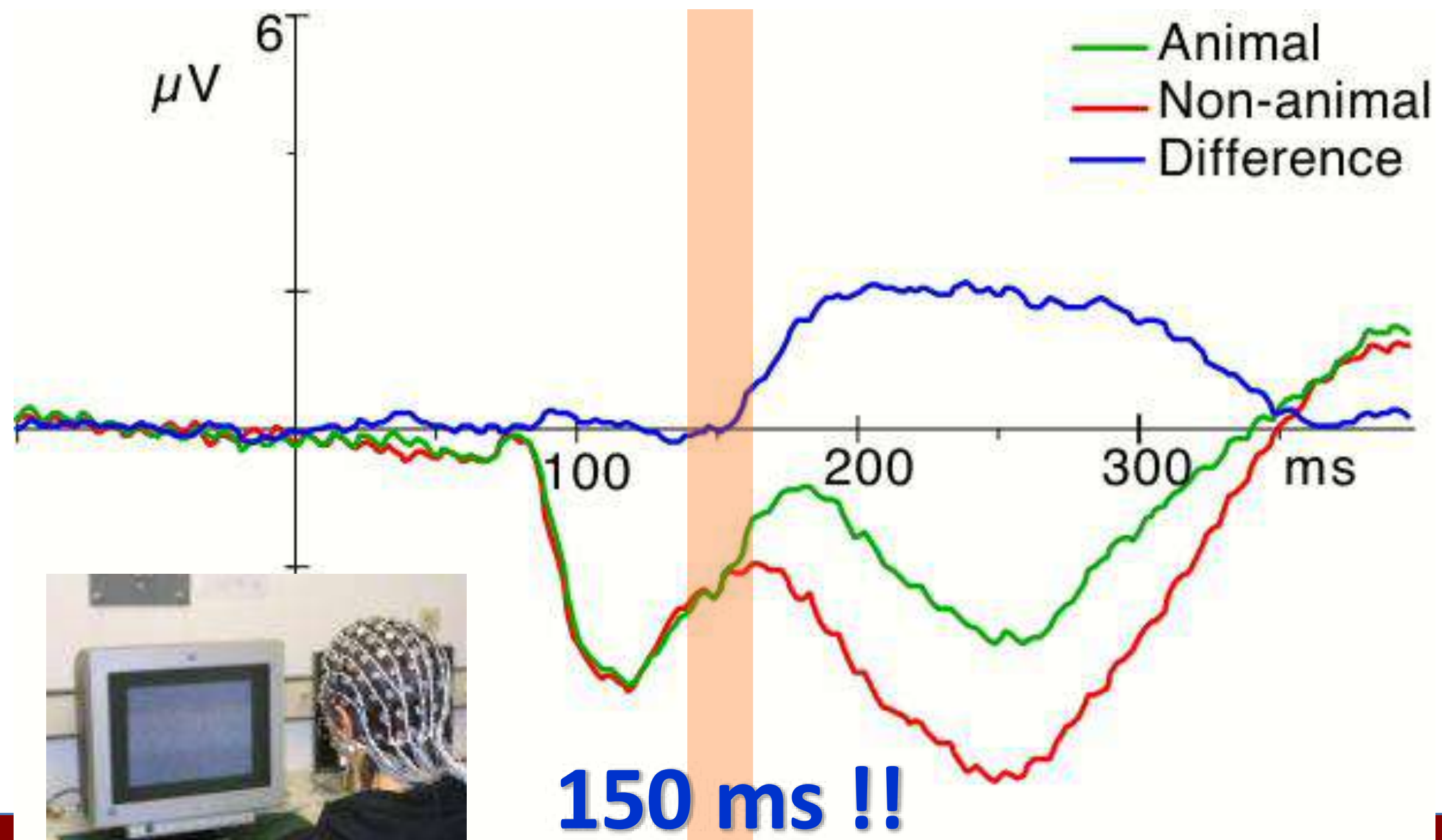


I. Biederman, *Science*, 1972



Speed of processing in the human visual system

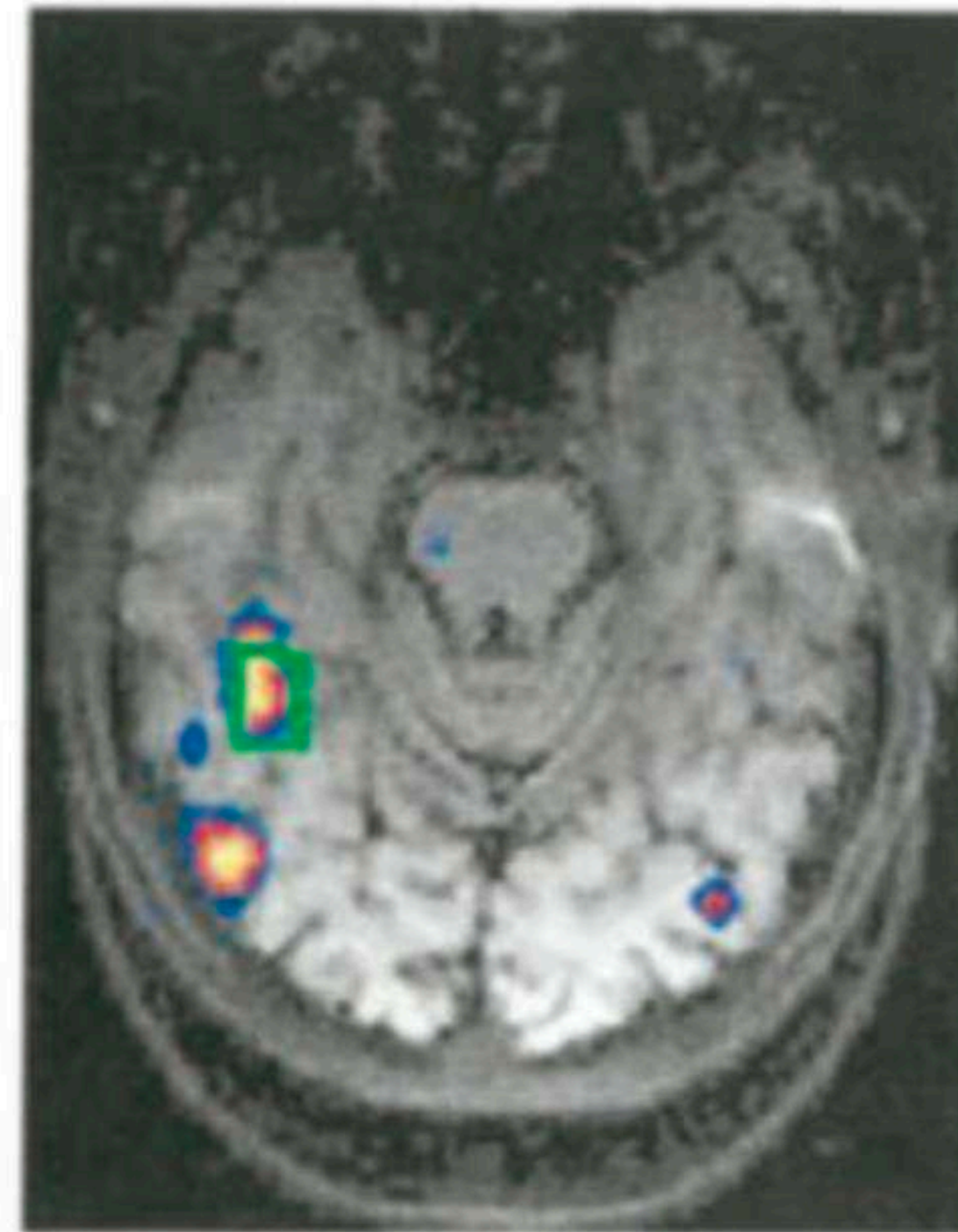
Simon Thorpe, Denis Fize & Catherine Marlot



Humans can recognize complex scenes very quickly.

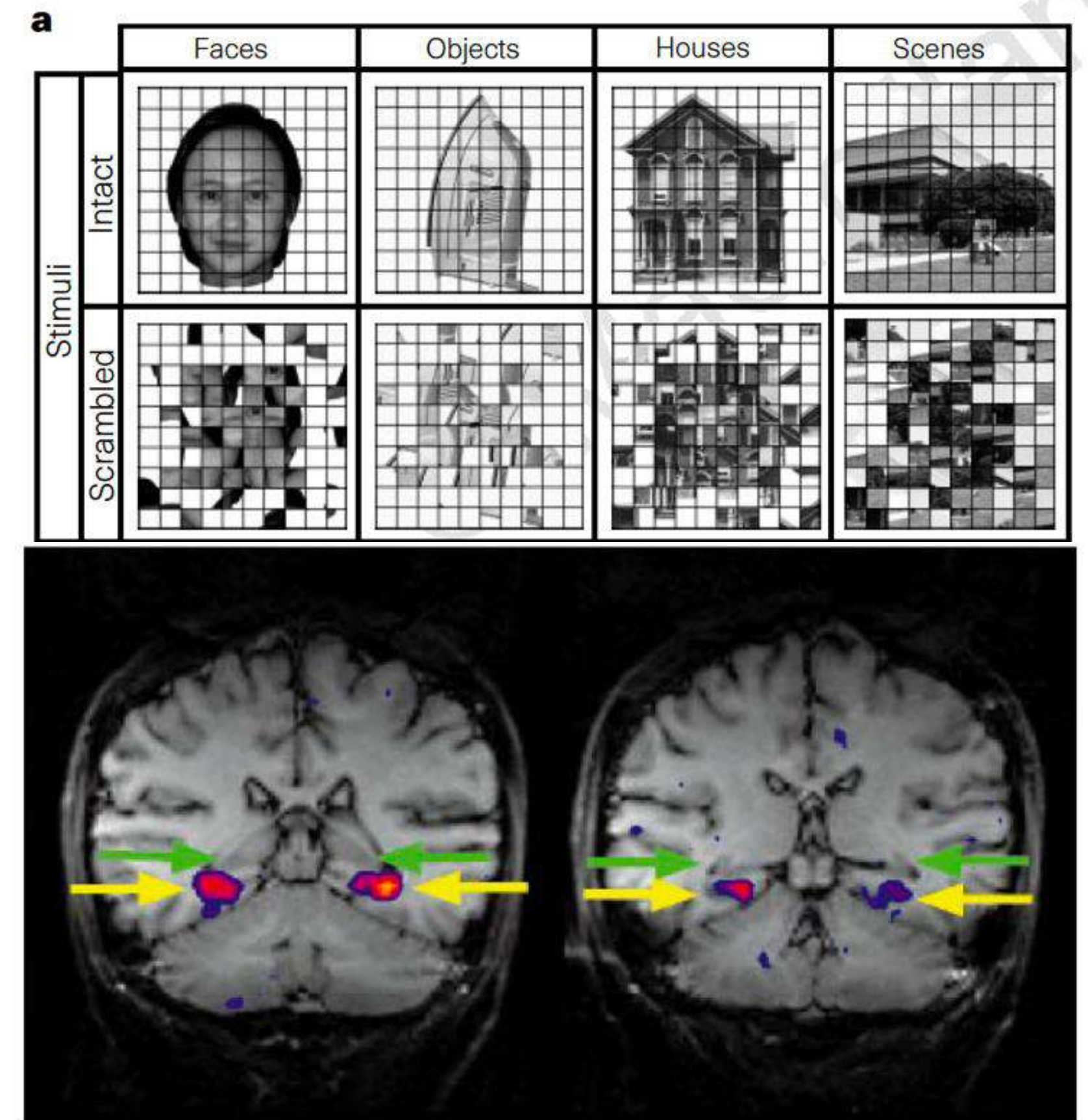
Neural correlates of object & scene recognition

Faces > Houses



% signal change

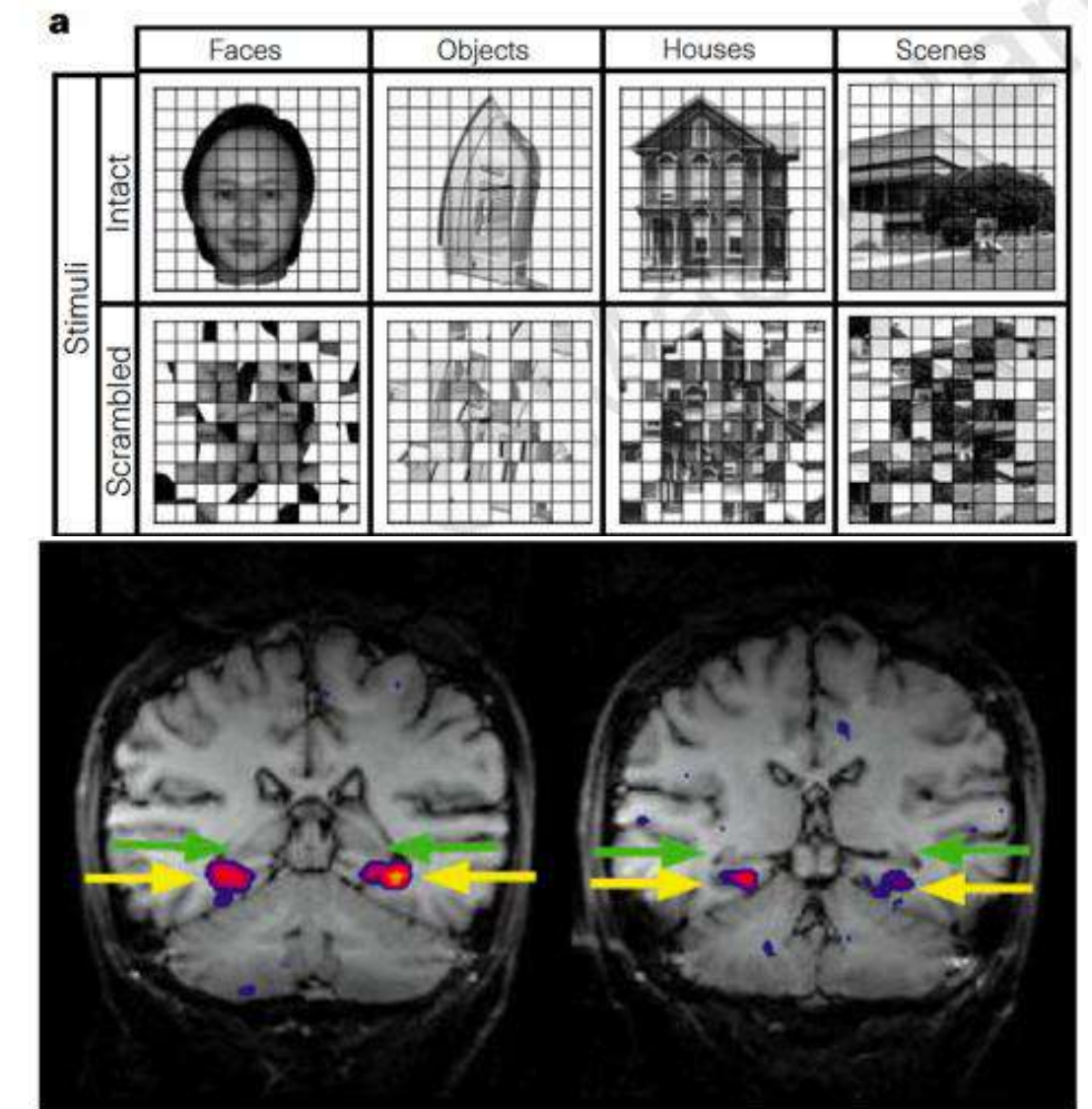
Kanwisher et al. J. Neuro. 1997



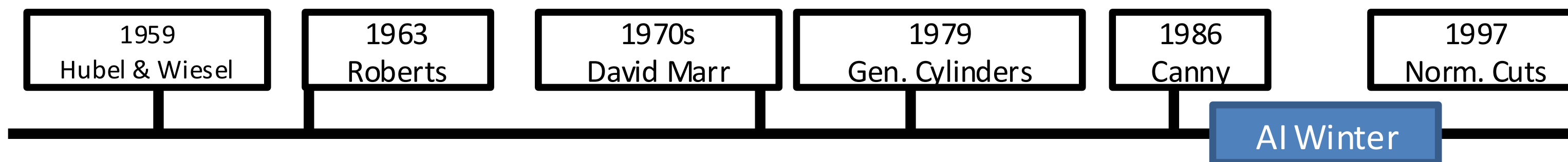
Epstein & Kanwisher, Nature, 1998

Different parts of the brain respond to different visual categories.

Visual recognition is a fundamental task for visual intelligence



Recognition via Grouping (1990s)



Normalized Cuts, Shi and Malik, 1997

how to group pixels into meaningful regions

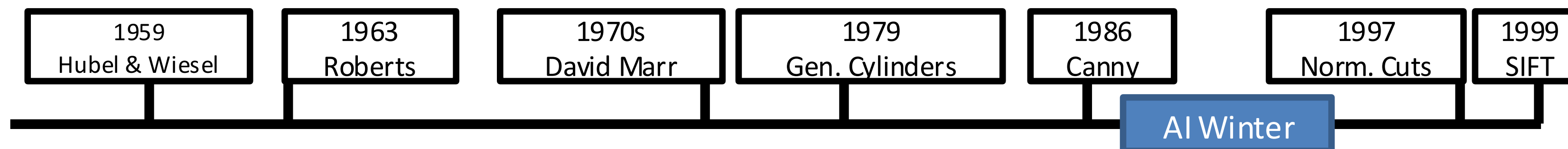
Recognition via Matching (2000s)



[Image](#) is public domain



[Image](#) is public domain



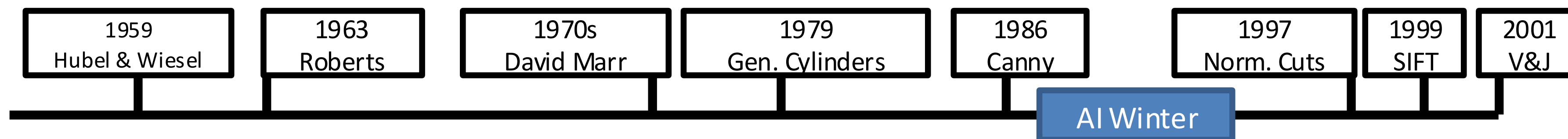
SIFT, David
Lowe, 1999

SIFT detects distinctive features that can be matched across different images, even when scale or viewpoint changes.

Face Detection

Viola and Jones, 2001

One of the first successful applications of machine learning to vision



Slide inspiration: Justin Johnson

Caltech 101 images

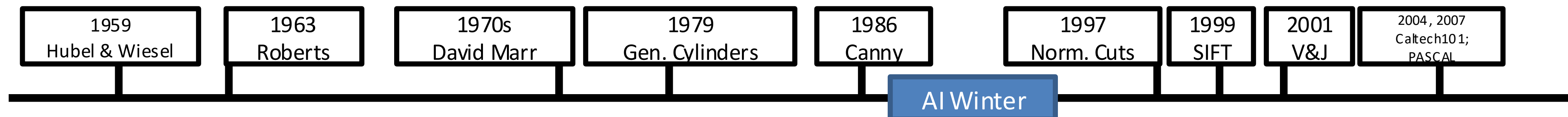


PASCAL Visual Object Challenge

Image is CC0 1.0 public domain



Image is CC0 1.0 public domain



The rise of standardized datasets and benchmarks

SUN, 131K images

[Xiao et al. **2010**]

LabelMe, 37K images

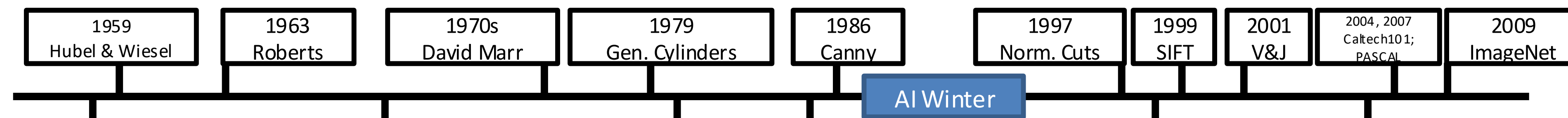
[Russell et al. **2007**]

PASCAL VOC, 30K images

[Everingham et al. **2006-2012**]

Caltech101, 9K images

[Fei-Fei, Fergus, Perona, **2003**]



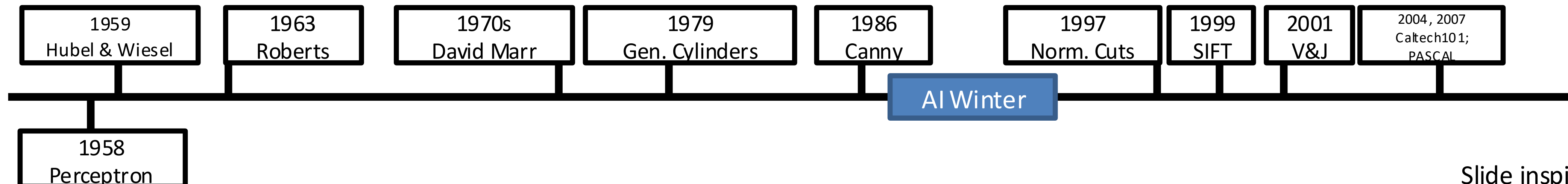
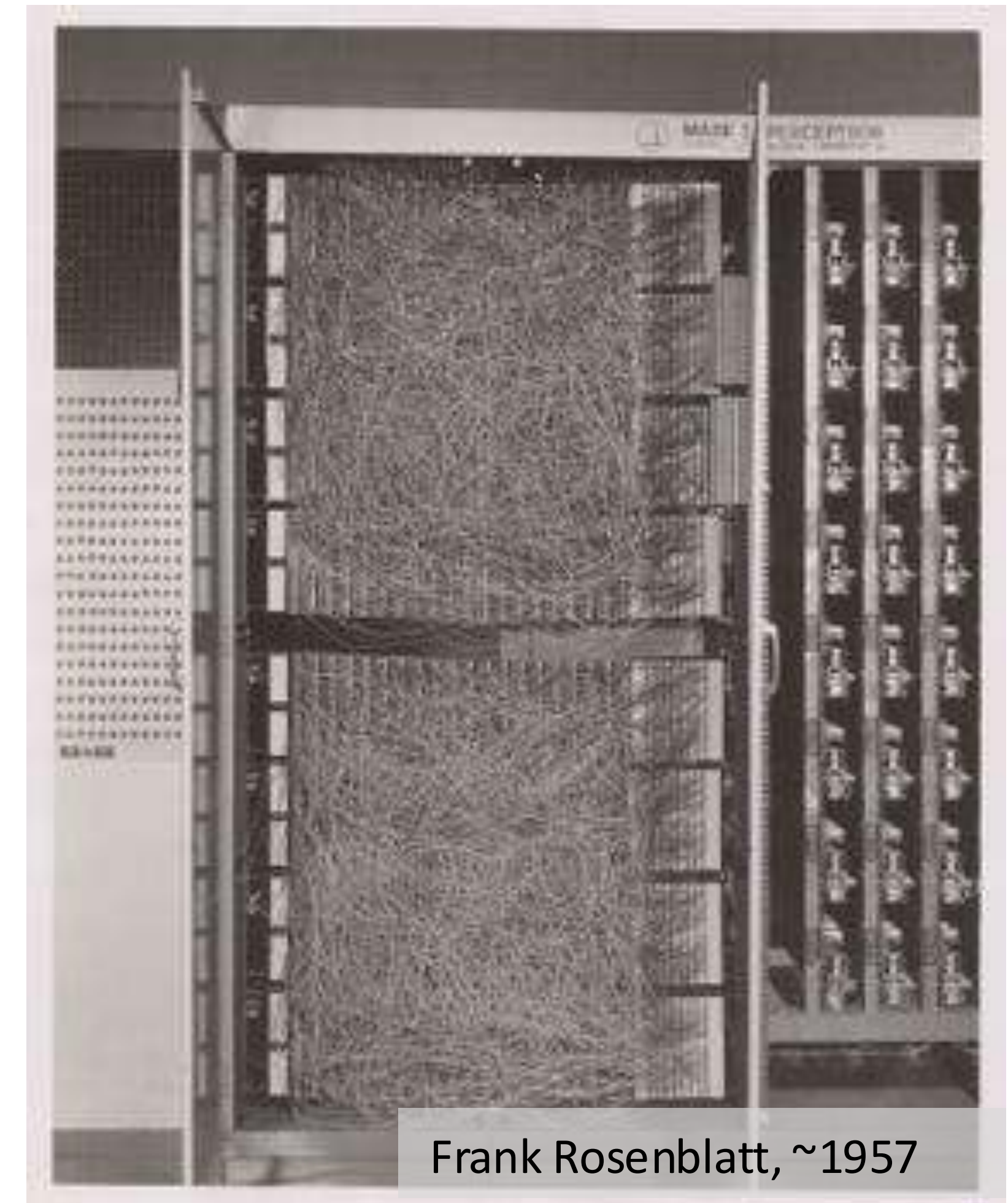
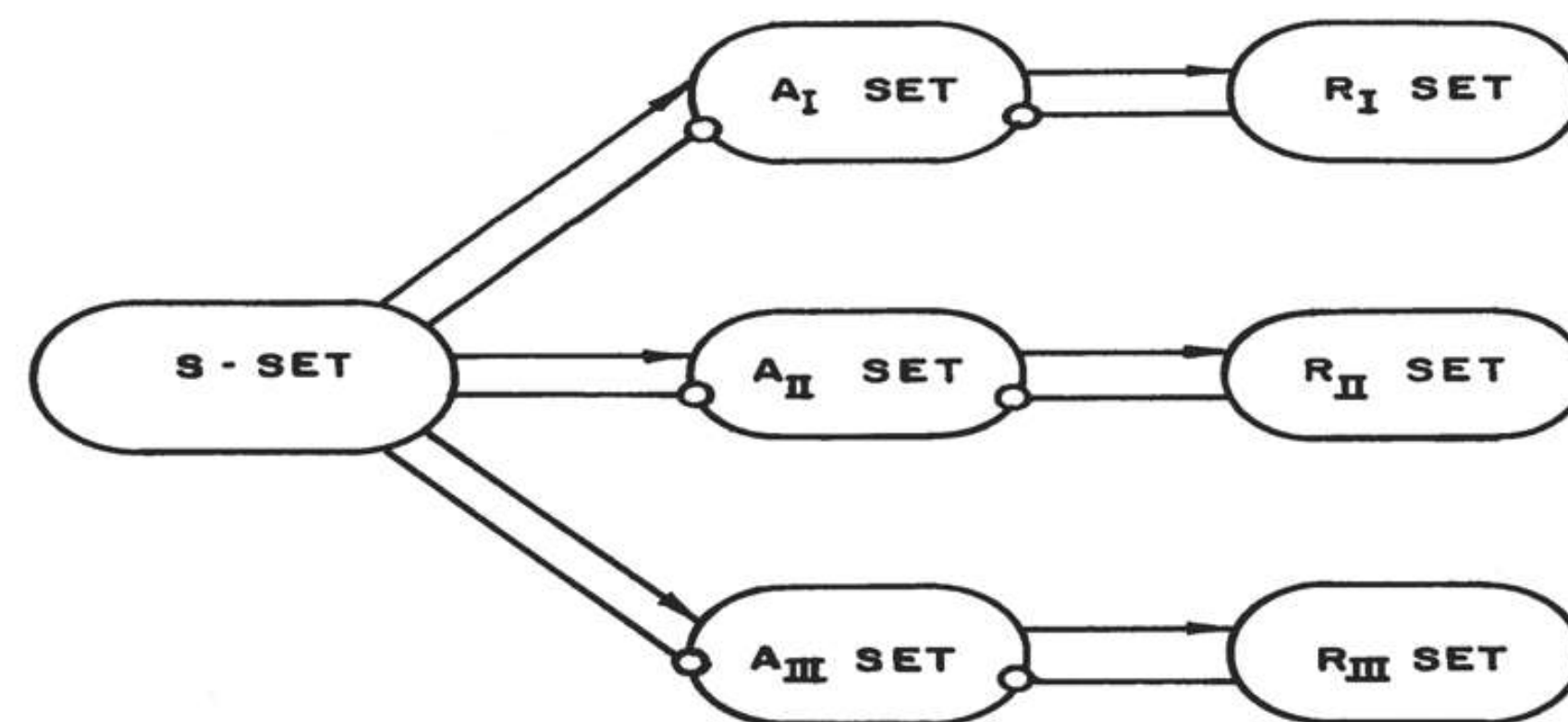
In parallel, neural networks were also developing.

Learning representations by back-propagating errors

David E. Rumelhart*, Geoffrey E. Hinton†
& Ronald J. Williams*

* Institute for Cognitive Science, C-015, University of California,
San Diego, La Jolla, California 92093, USA

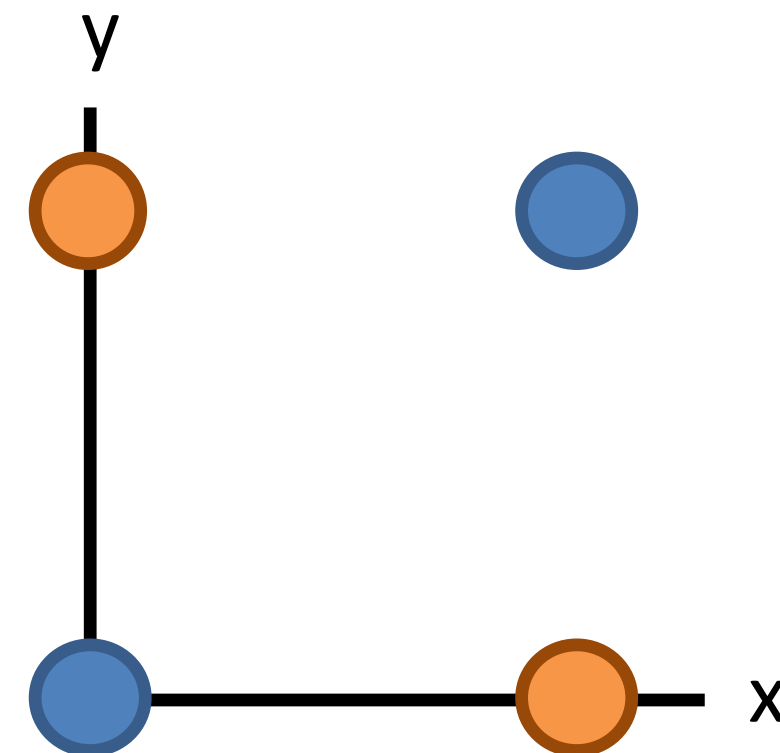
† Department of Computer Science, Carnegie-Mellon University,
Pittsburgh, Philadelphia 15213, USA



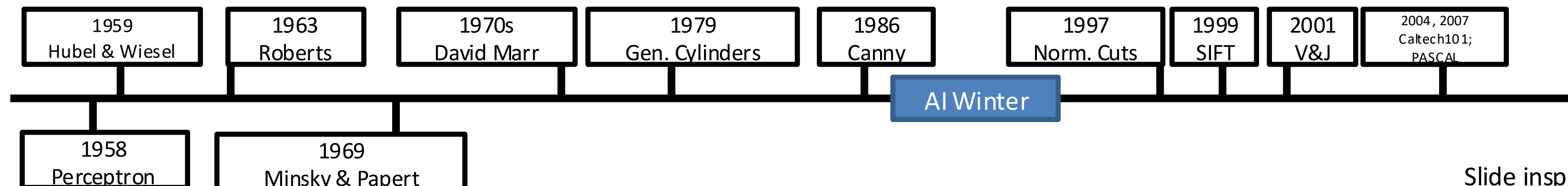
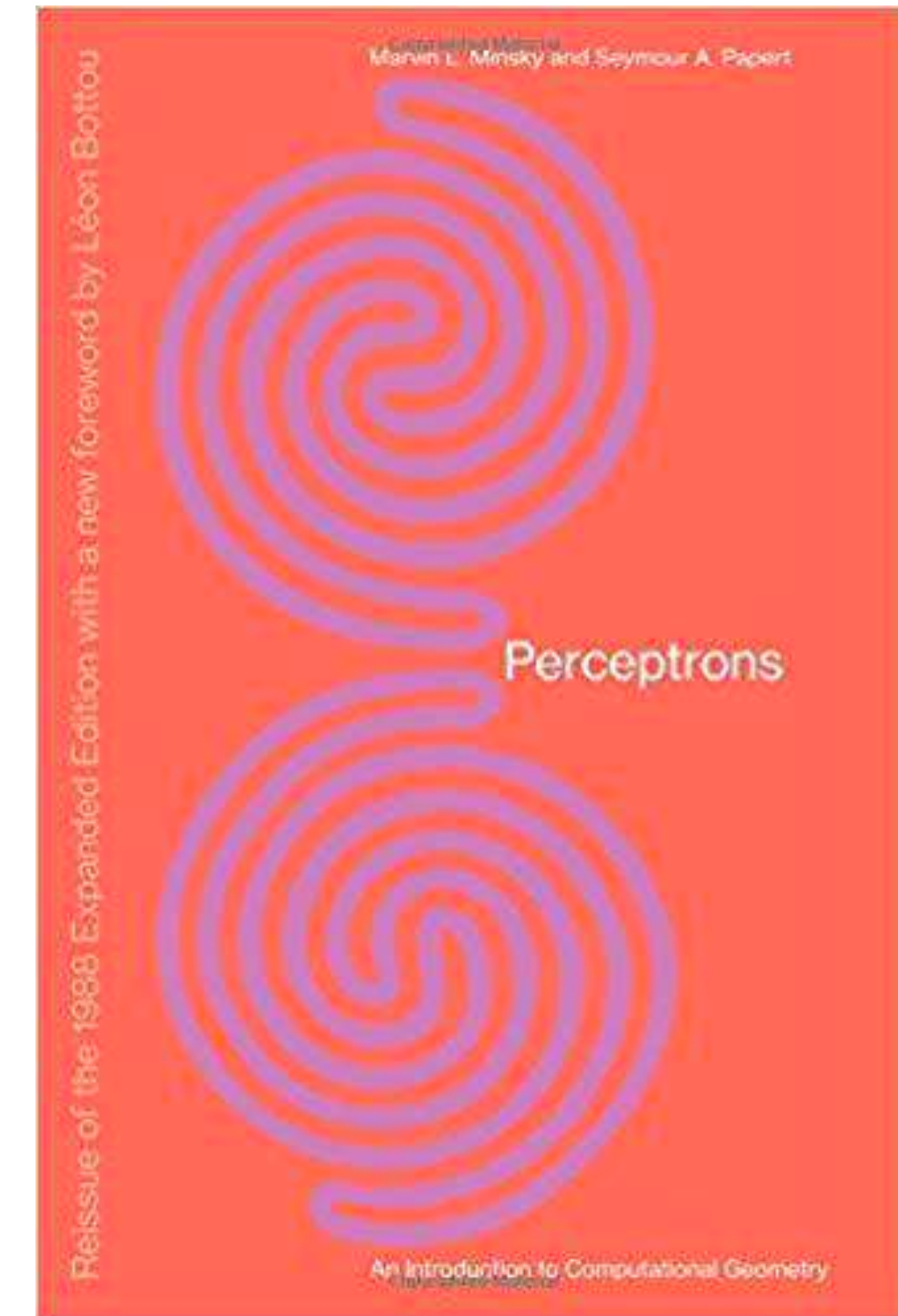
Slide inspiration: Justin Johnson

Minsky and Papert, 1969

X	Y	F(x,y)
0	0	0
0	1	1
1	0	1
1	1	0



Showed that Perceptrons could not learn the XOR function
Caused a lot of disillusionment in the field



Slide inspiration: Justin Johnson

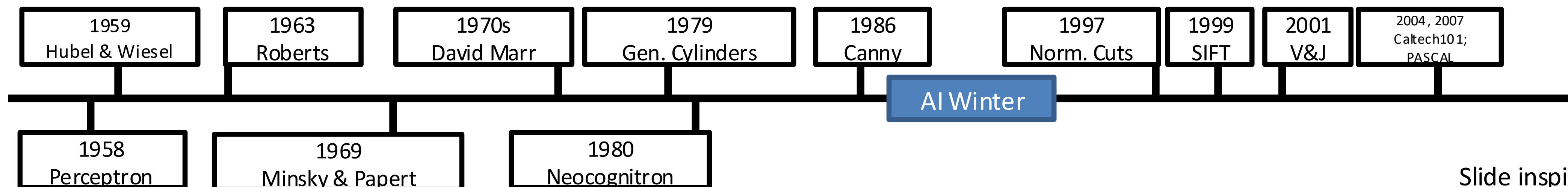
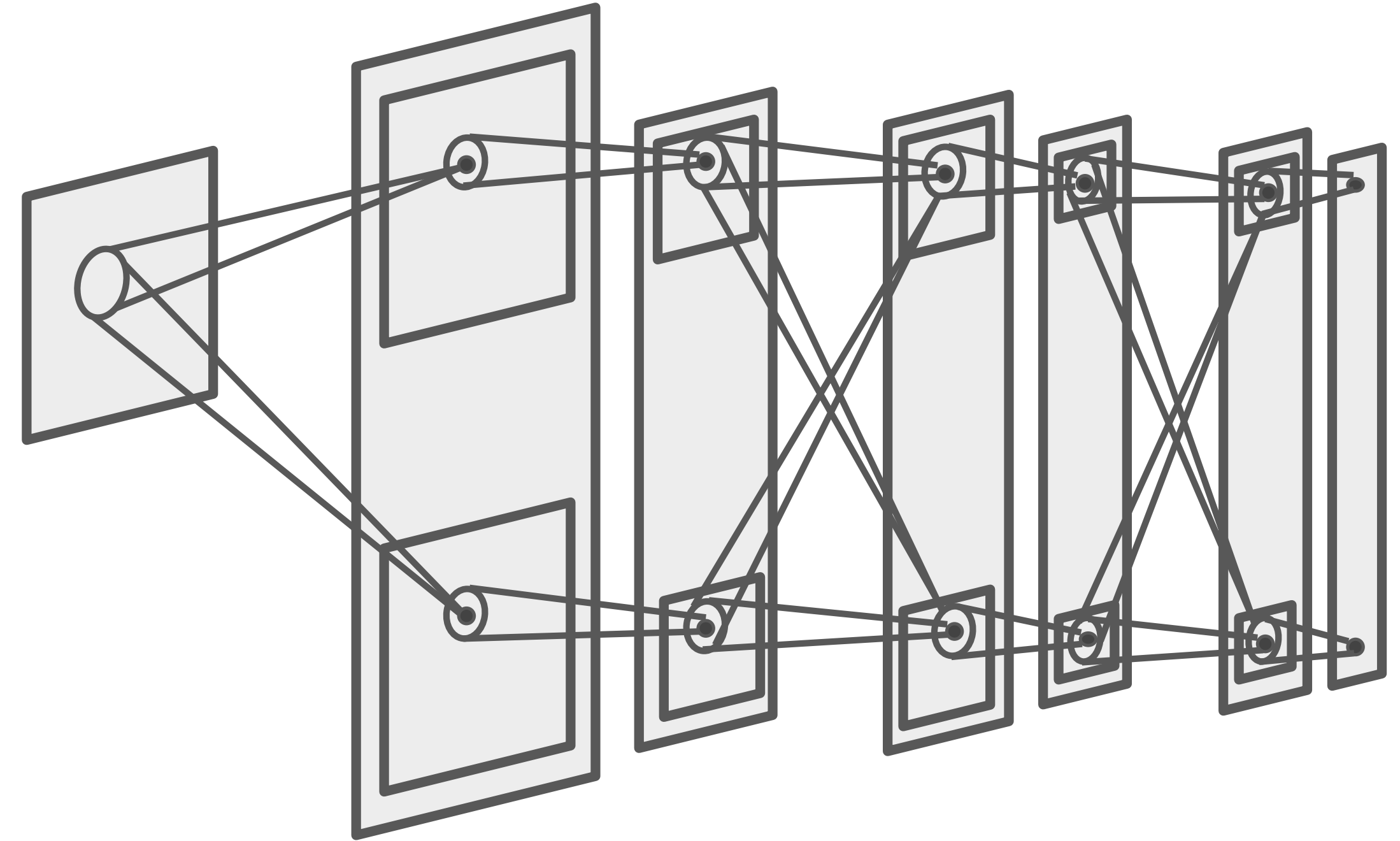
But a single-layer perceptron has clear limitations.

Neocognitron: Fukushima, 1980

Computational model the visual system,
directly inspired by Hubel and Wiesel's
hierarchy of complex and simple cells

Interleaved simple cells (convolution)
and complex cells (pooling)

No practical training algorithm

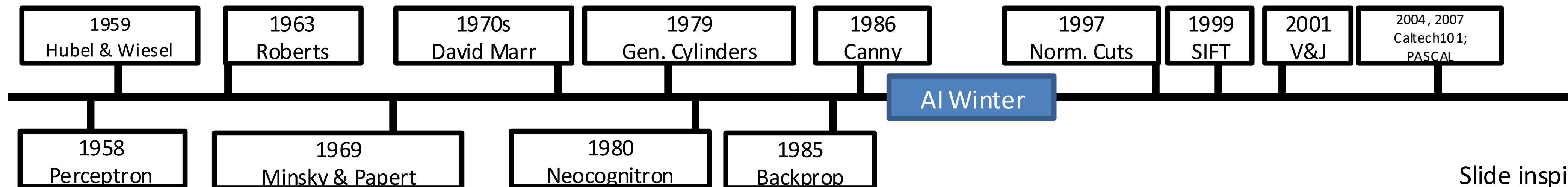
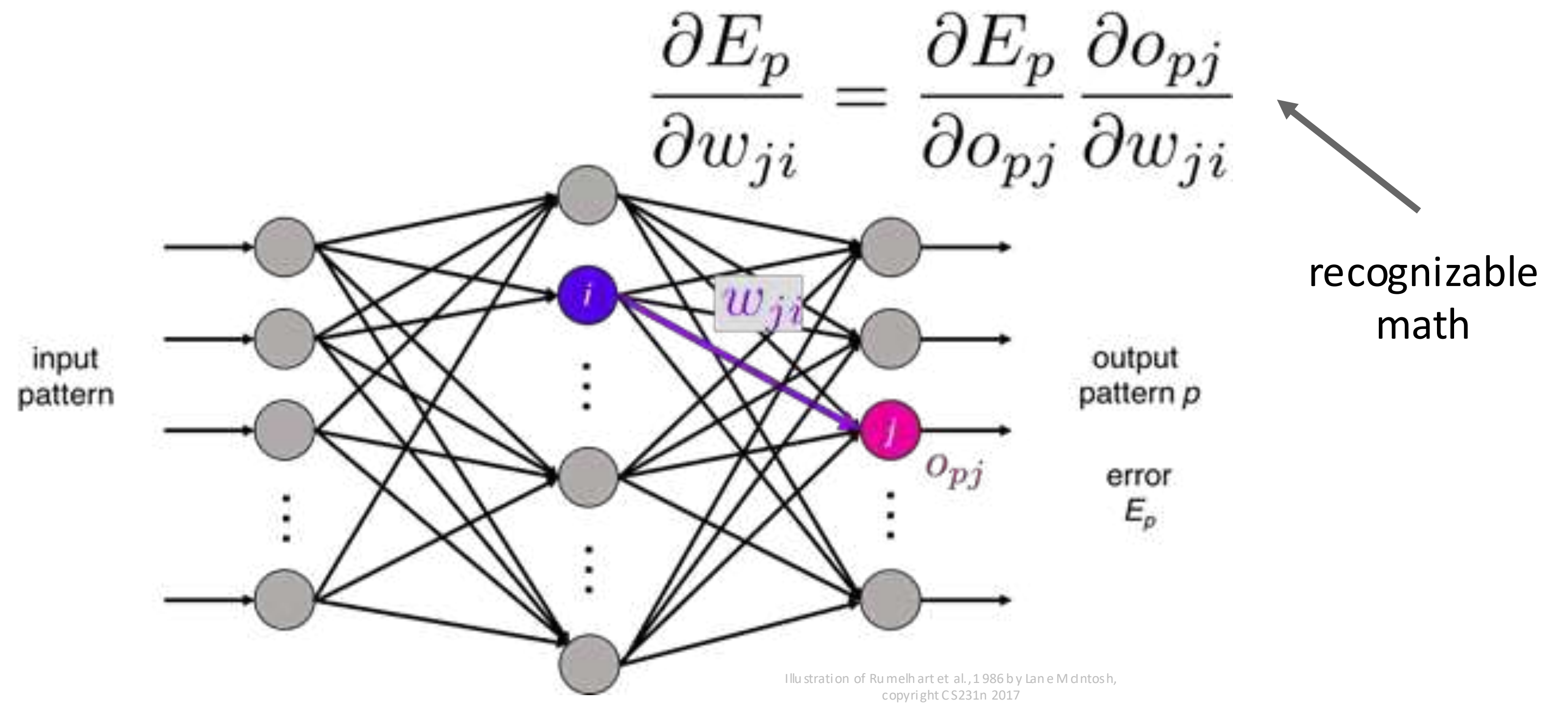


Slide inspiration: Justin Johnson

Backprop: Rumelhart, Hinton, and Williams, 1986

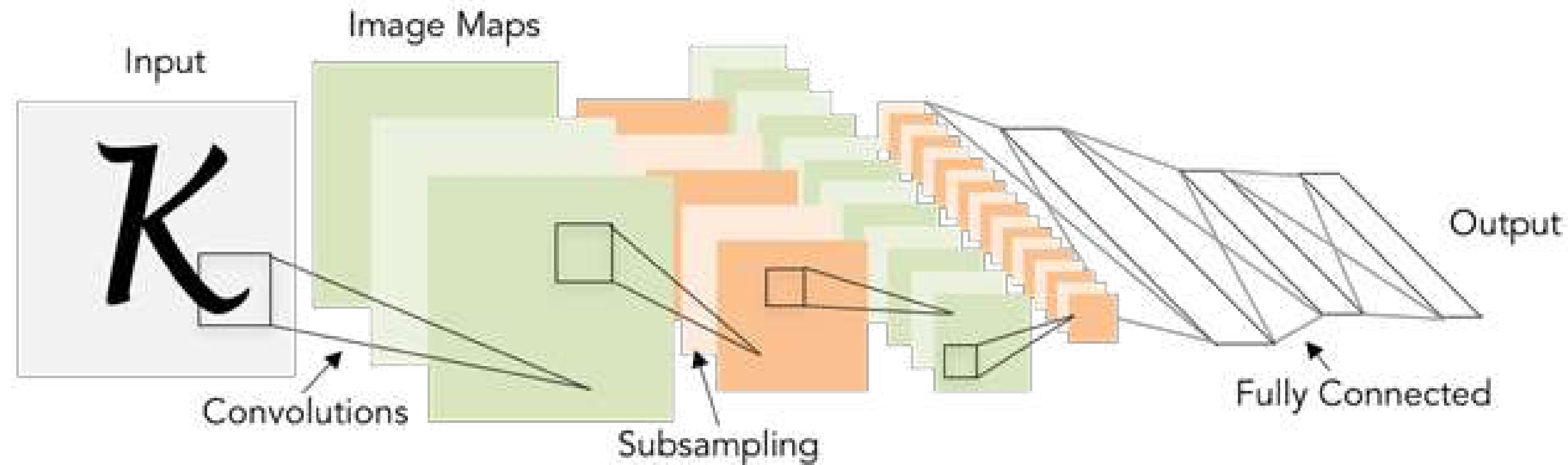
Introduced
backpropagation
for computing
gradients in neural
networks

Successfully trained
perceptrons with
multiple layers



Slide inspiration: Justin Johnson

Convolutional Networks: LeCun et al, 1998

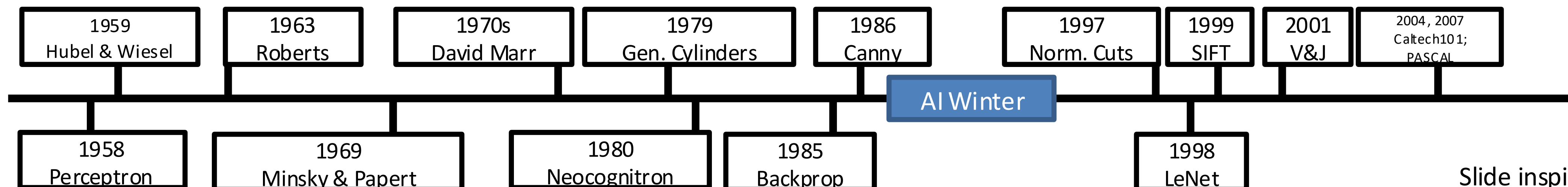


Applied backprop algorithm to a Neocognitron-like architecture

Learned to recognize handwritten digits

Was deployed in a commercial system by NEC, processed handwritten checks

Very similar to our modern convolutional networks!



Slide inspiration: Justin Johnson

2000s: “Deep Learning”

People tried to train neural networks that were deeper and deeper

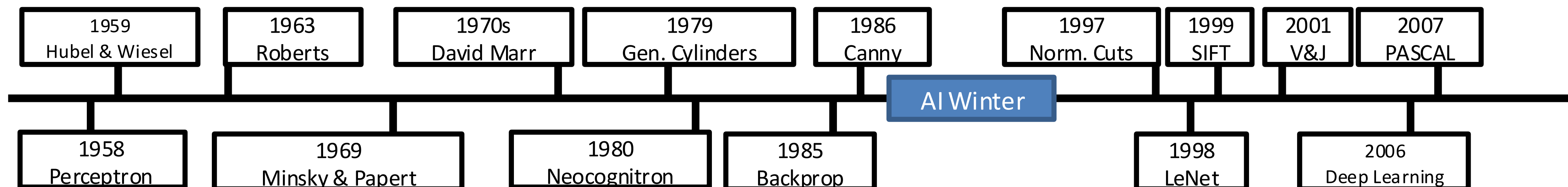
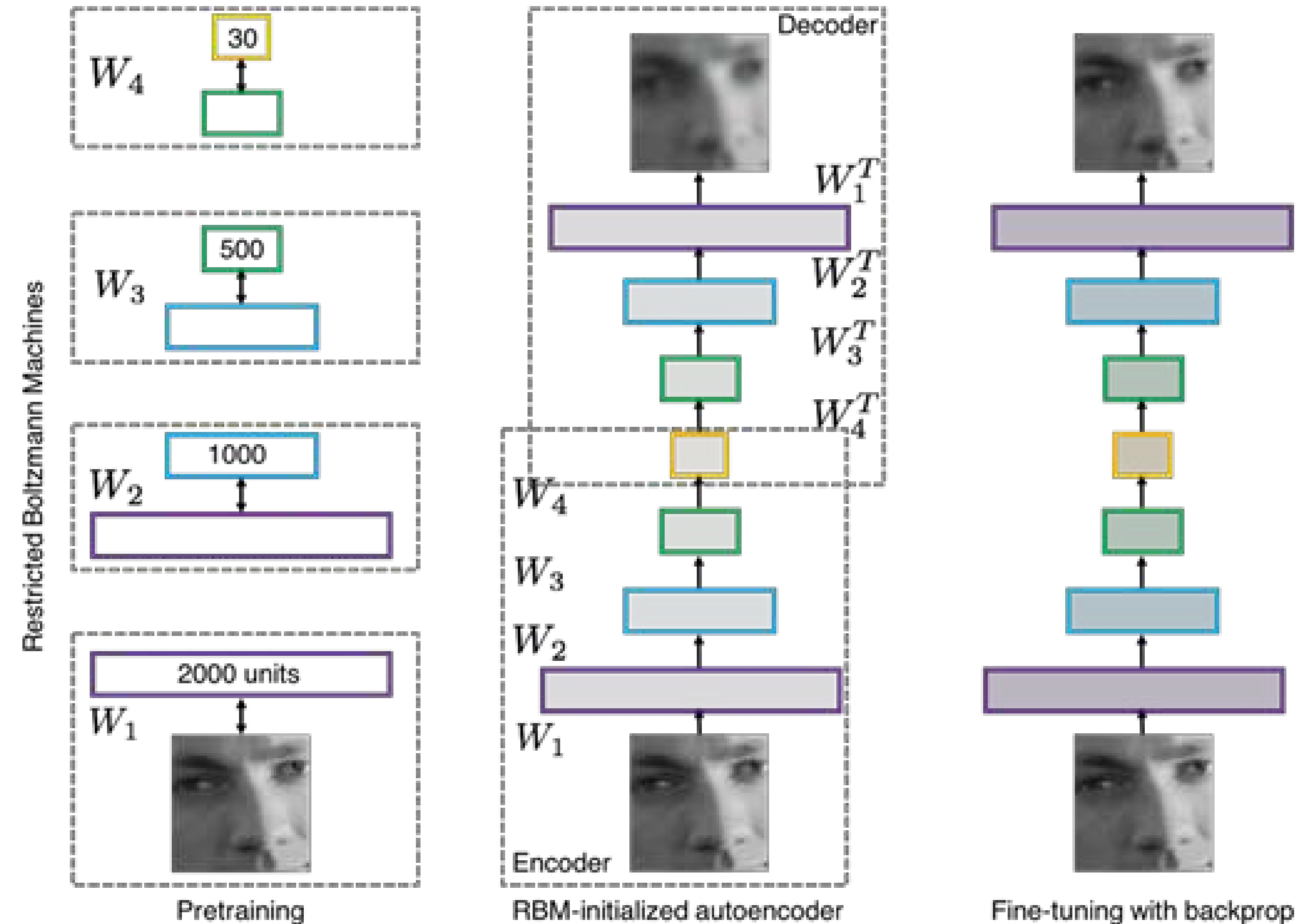
Not a mainstream research topic at this time

Hinton and Salakhutdinov, 2006

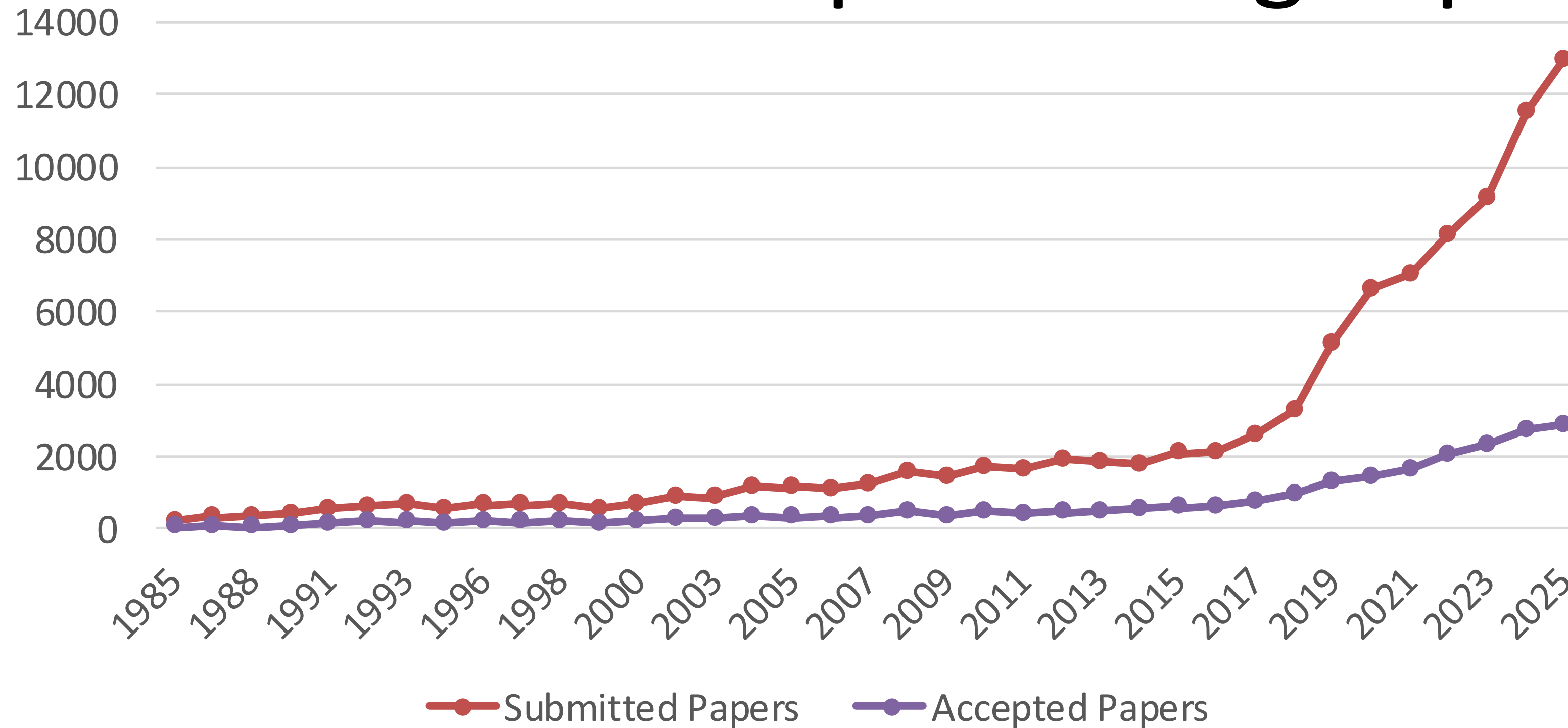
Bengio et al, 2007

Lee et al, 2009

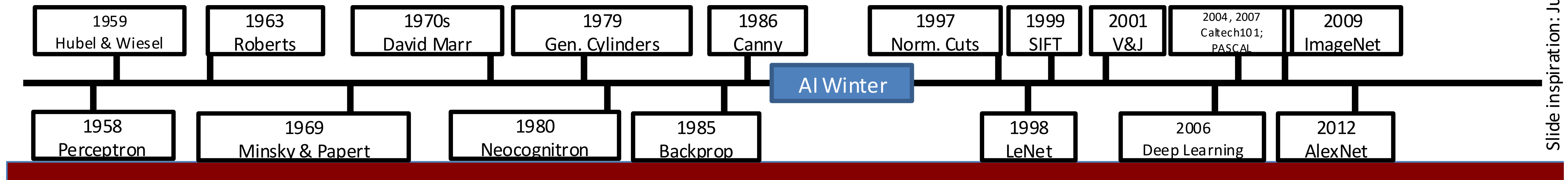
Glorot and Bengio, 2010



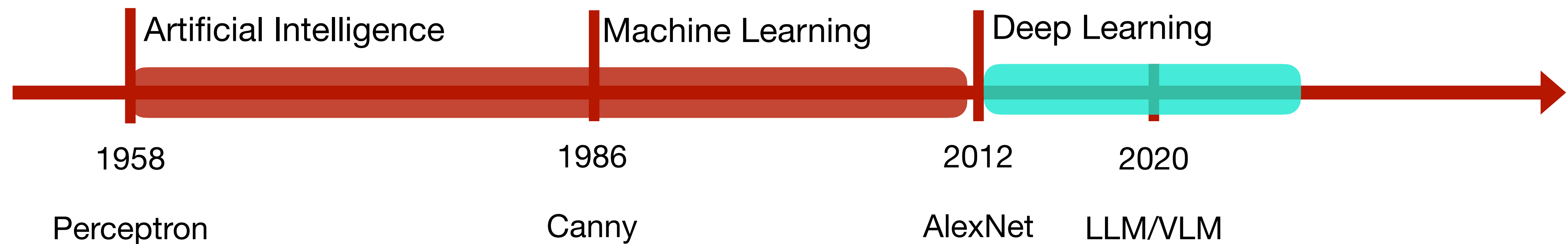
2012 to Present: Deep Learning Explosion



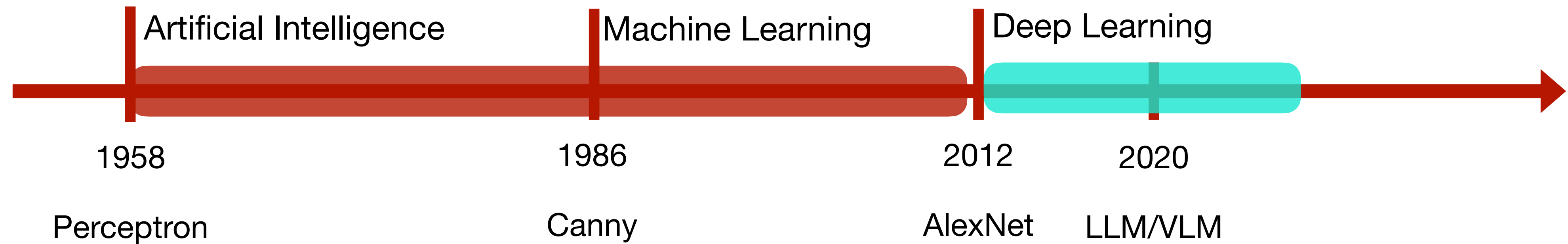
Publications at top Computer Vision conference



Before 2012: Classical Computer Vision



After 2012: Deep Learning for Computer Vision

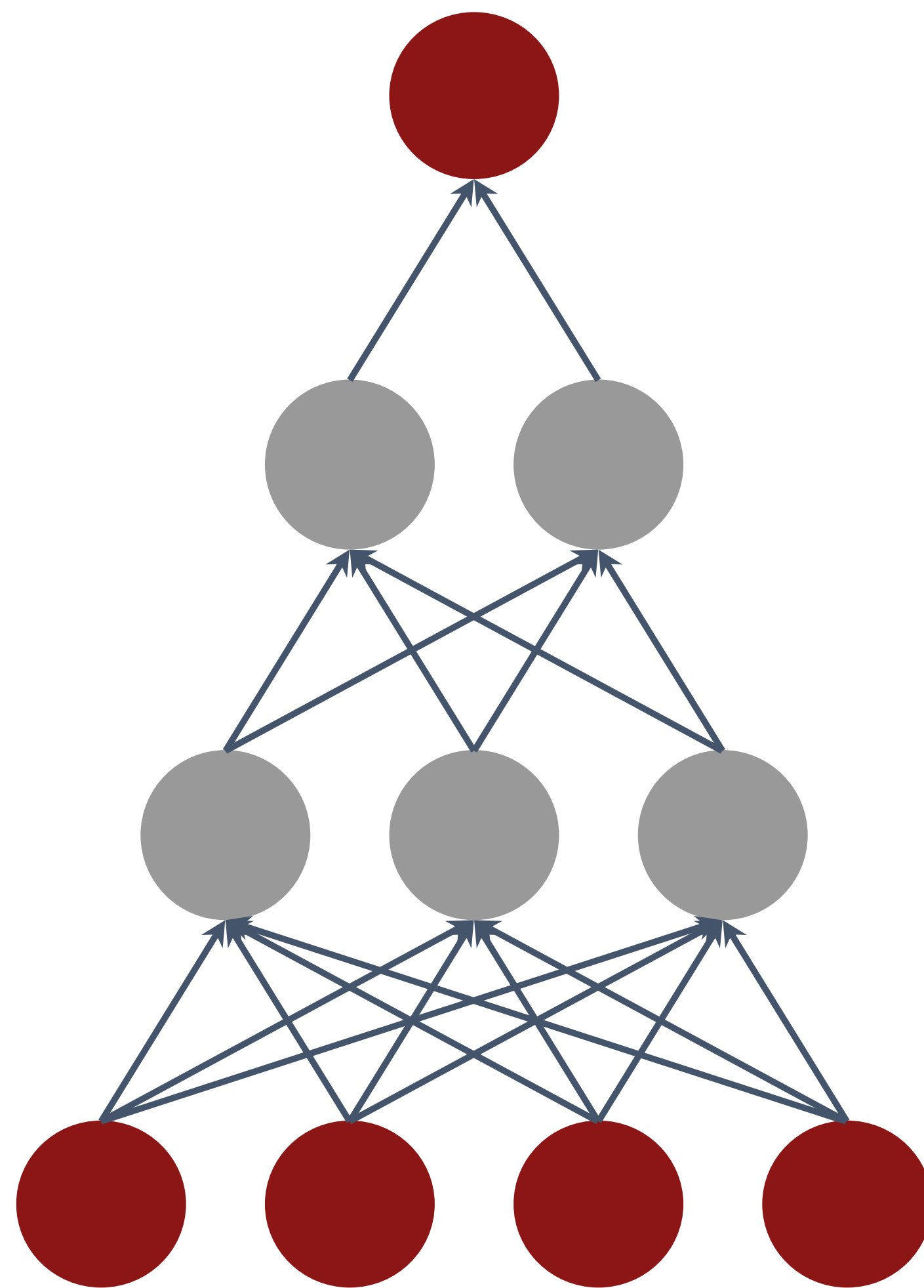




Data



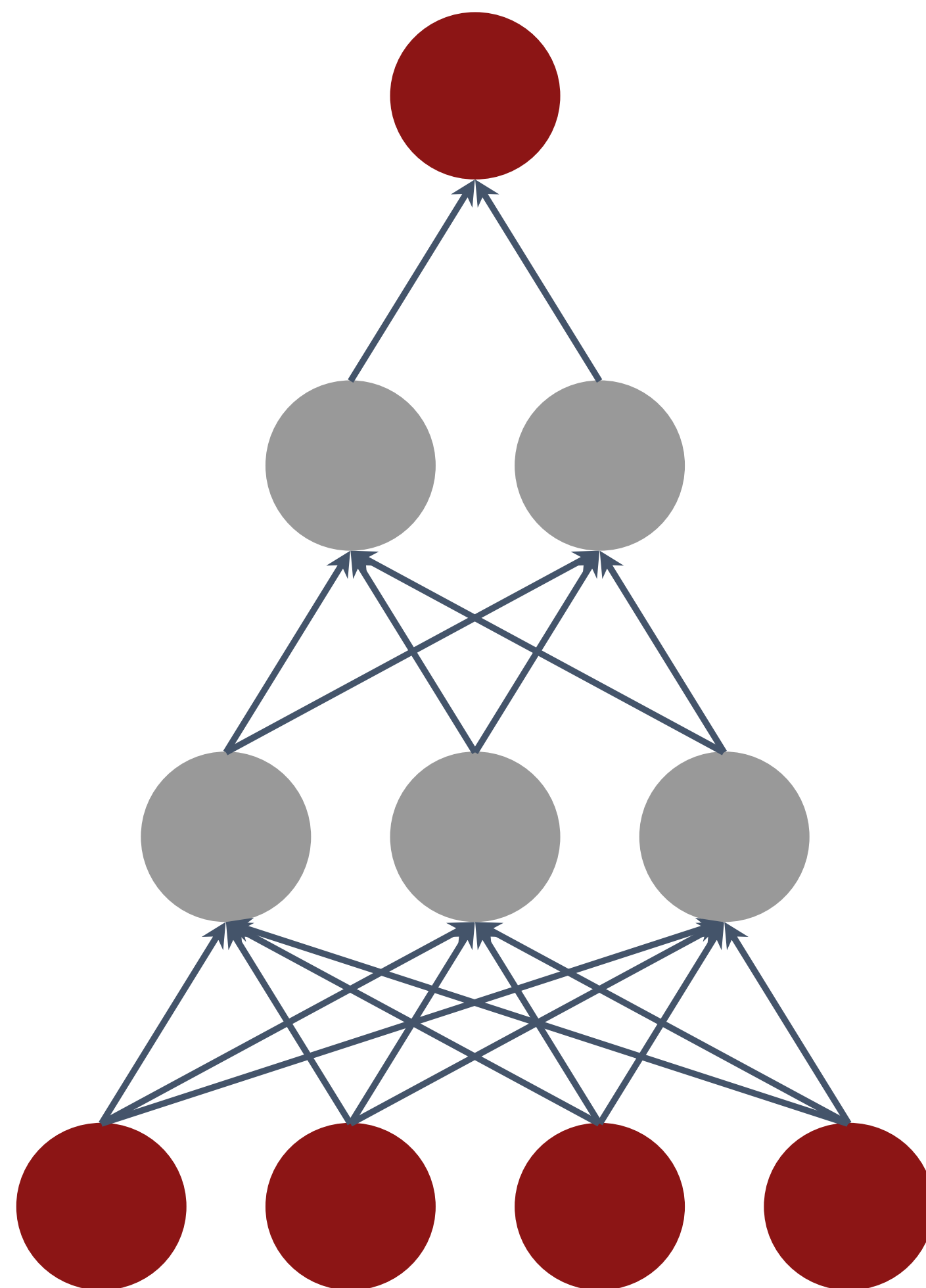
Data



Algorithms



Data



Algorithms

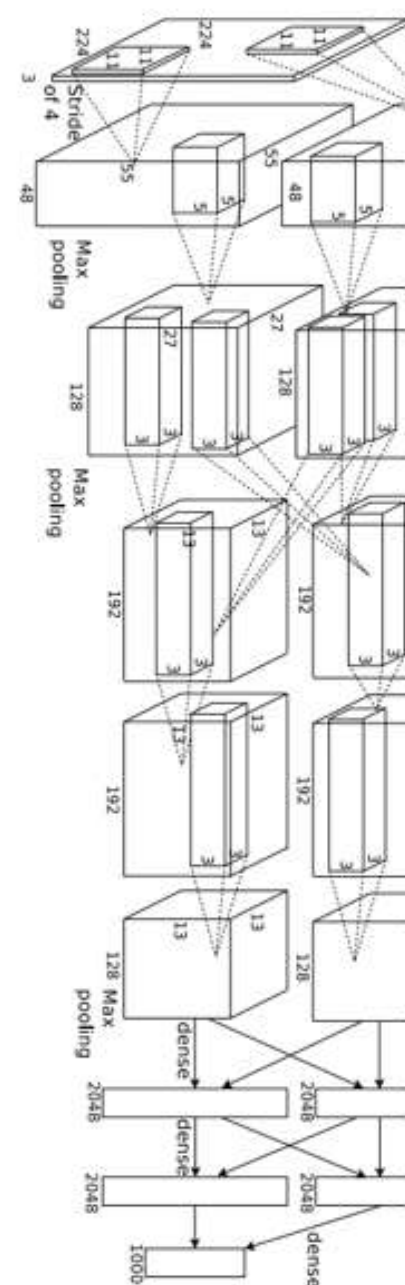


Computation

2012 to Present: Deep Learning is Everywhere

Year 2012

AlexNet

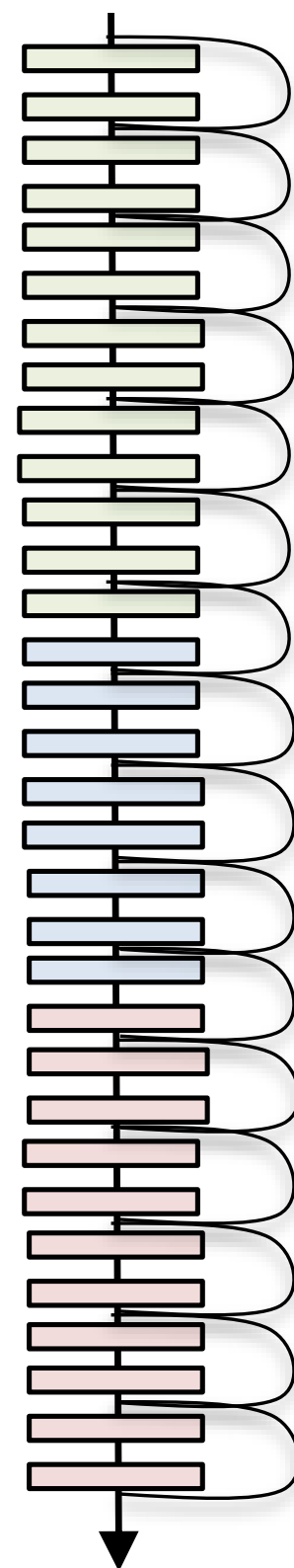


[Krizhevsky NIPS 2012]

Figure copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

Year 2015

ResNet

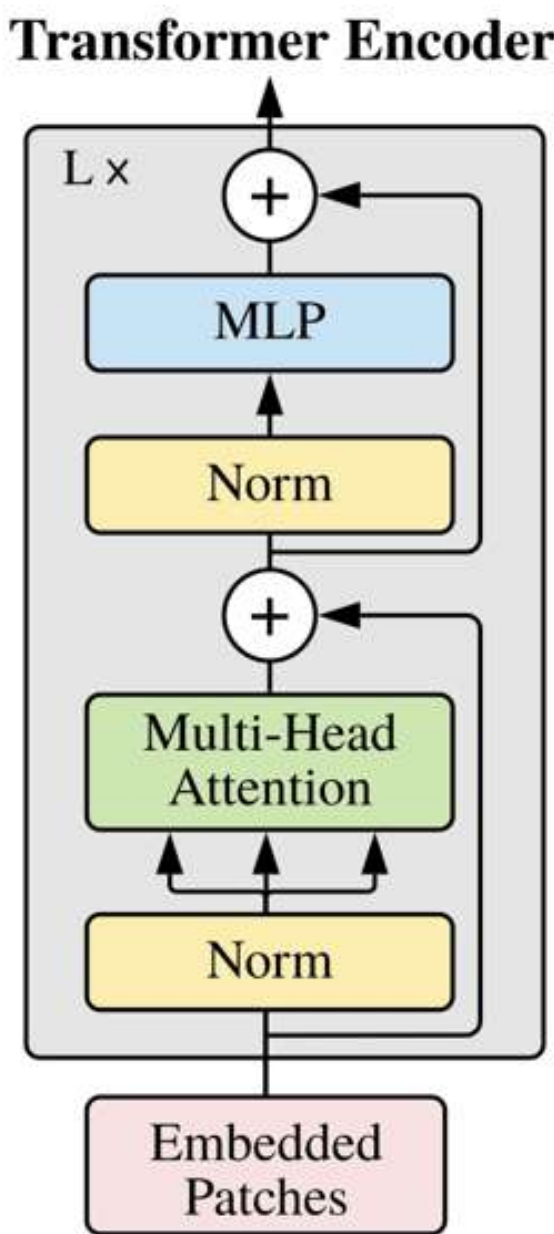


[He ICCV 2015]

CNNs

Year 2021

Vision Transformer (ViT)

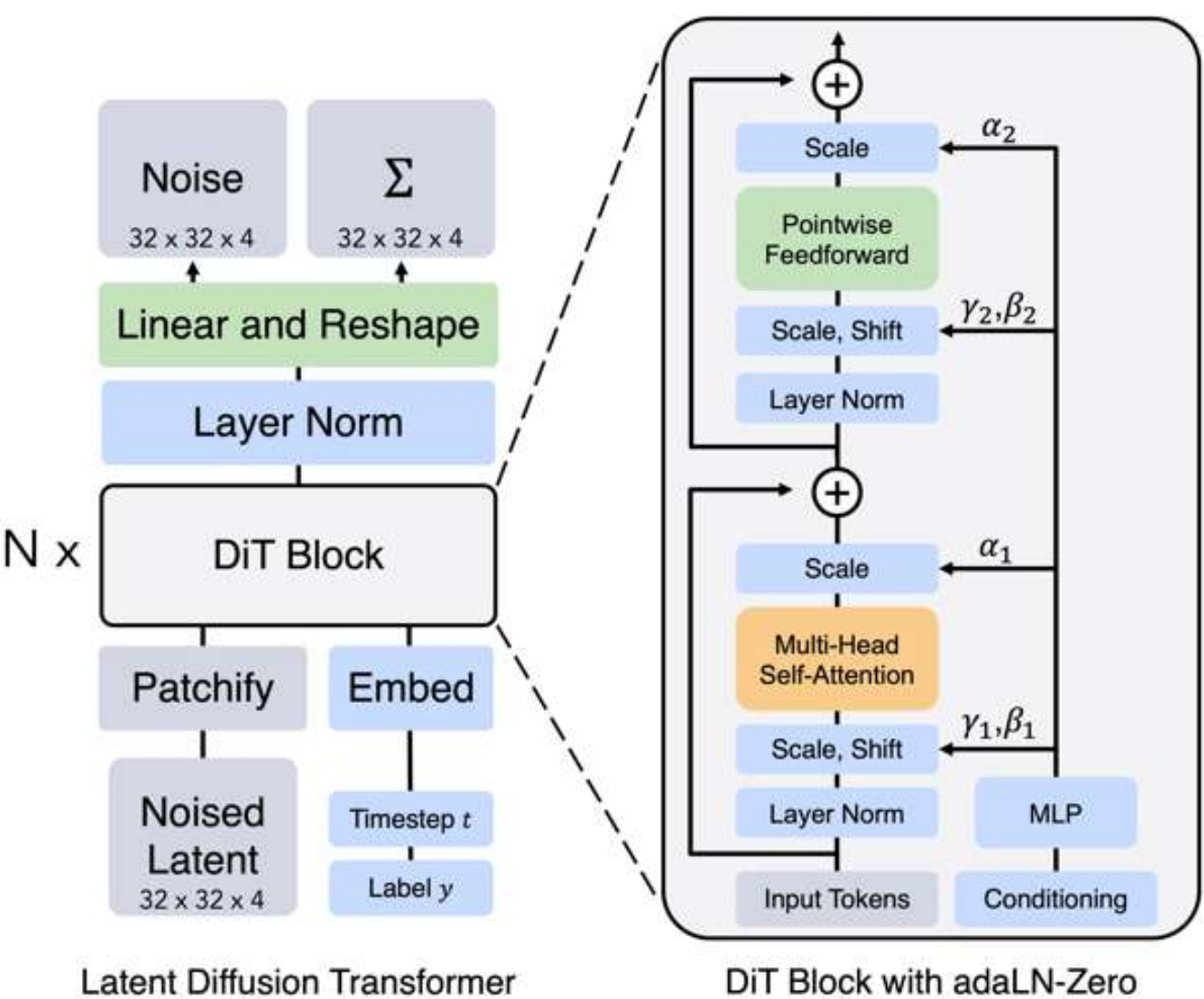


[Dosovitskiy ICLR 2021]

Transformers

Year 2023

Diffusion Transformer (DiT)



[Peebles ICCV 2023]

2012 to Present: Deep Learning is Everywhere

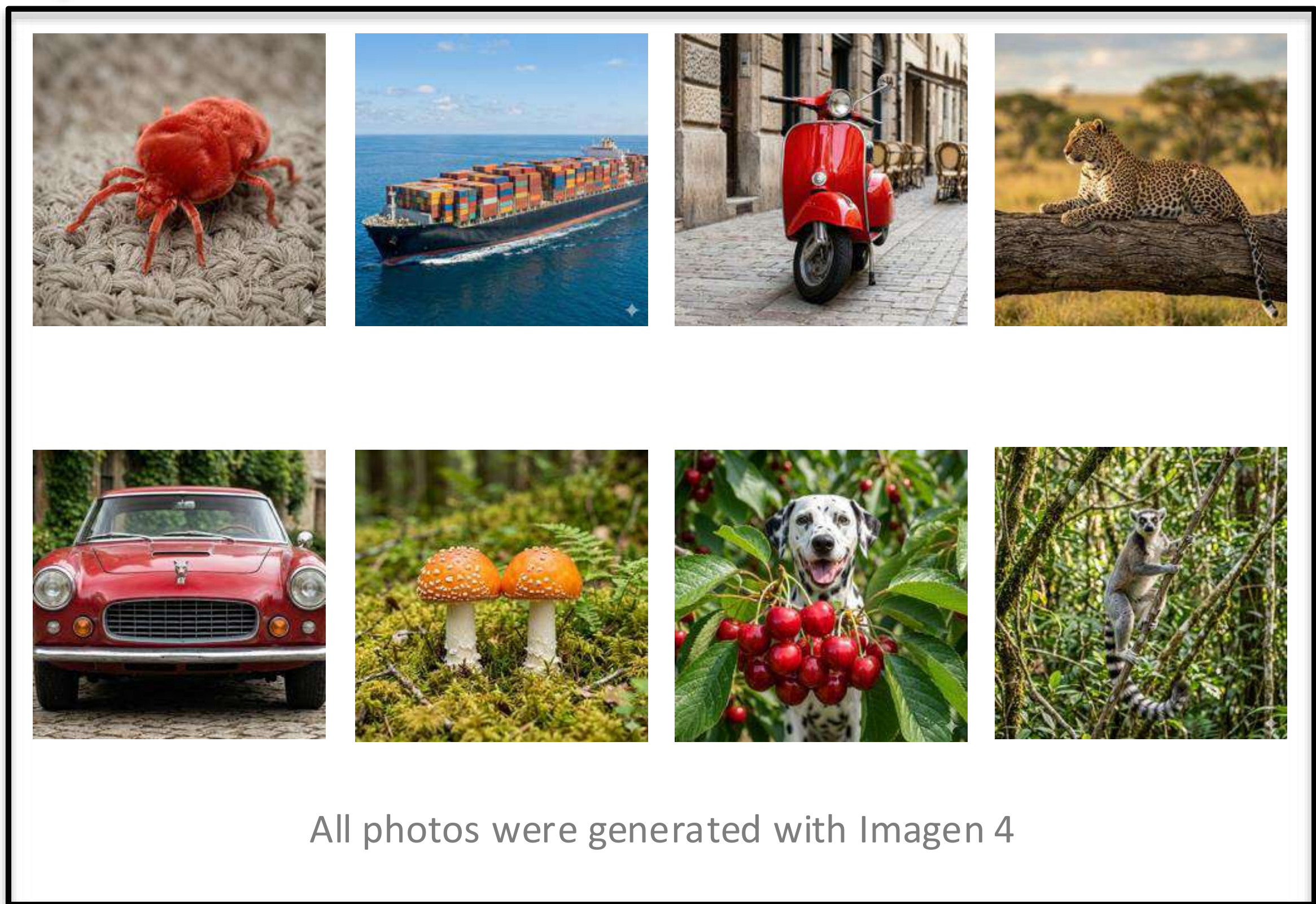
Image Classification (circa 2012)



Image Generation (2026)



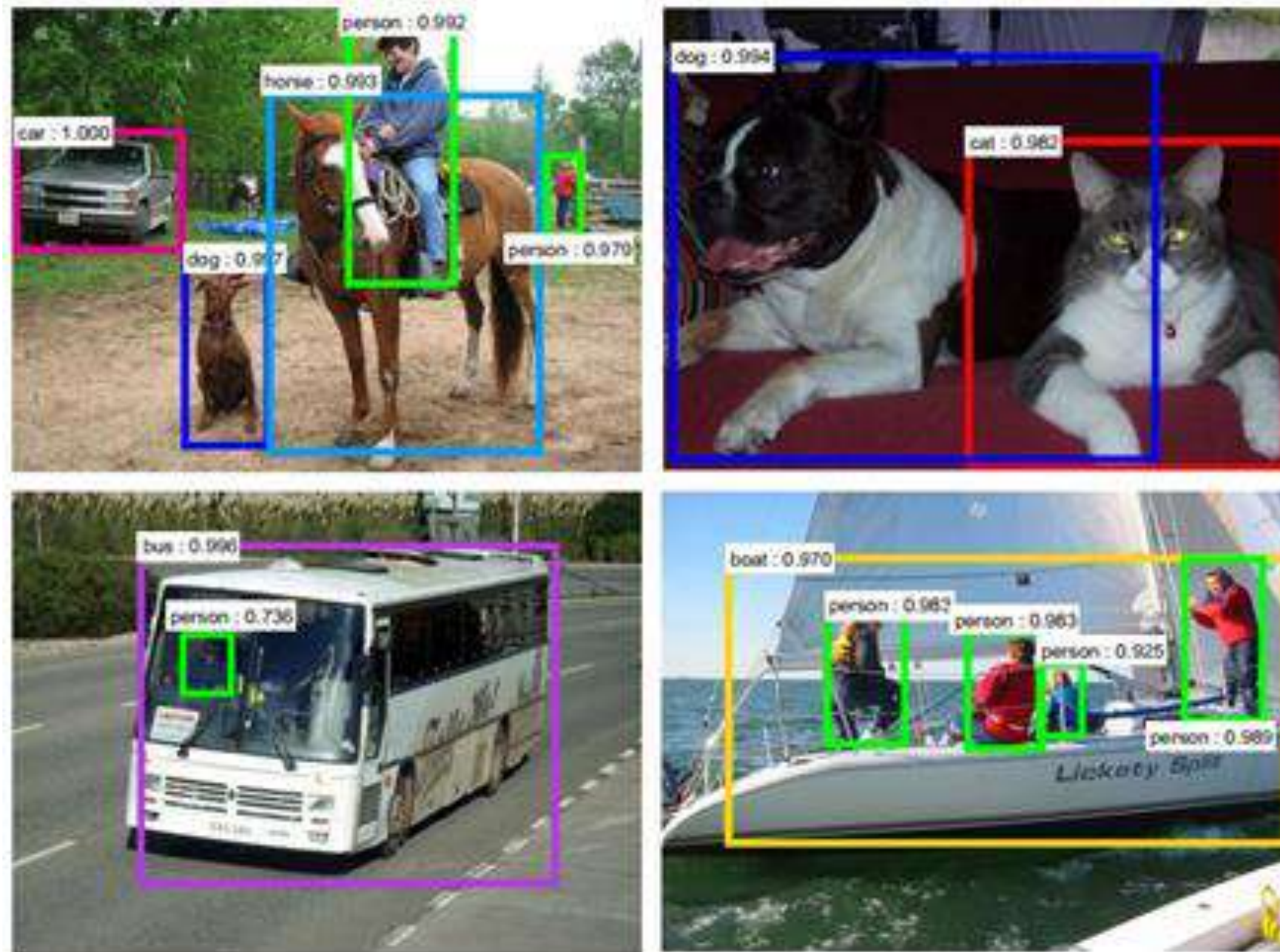
Figures copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.



Figures copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

2012 to Present: Deep Learning is Everywhere

Object Detection



Faster R-CNN
Ren, He, Girshick, and Sun, NeurIPS 2016

Image Segmentation



Segment Anything
Kirillov et al, ICCV 2023

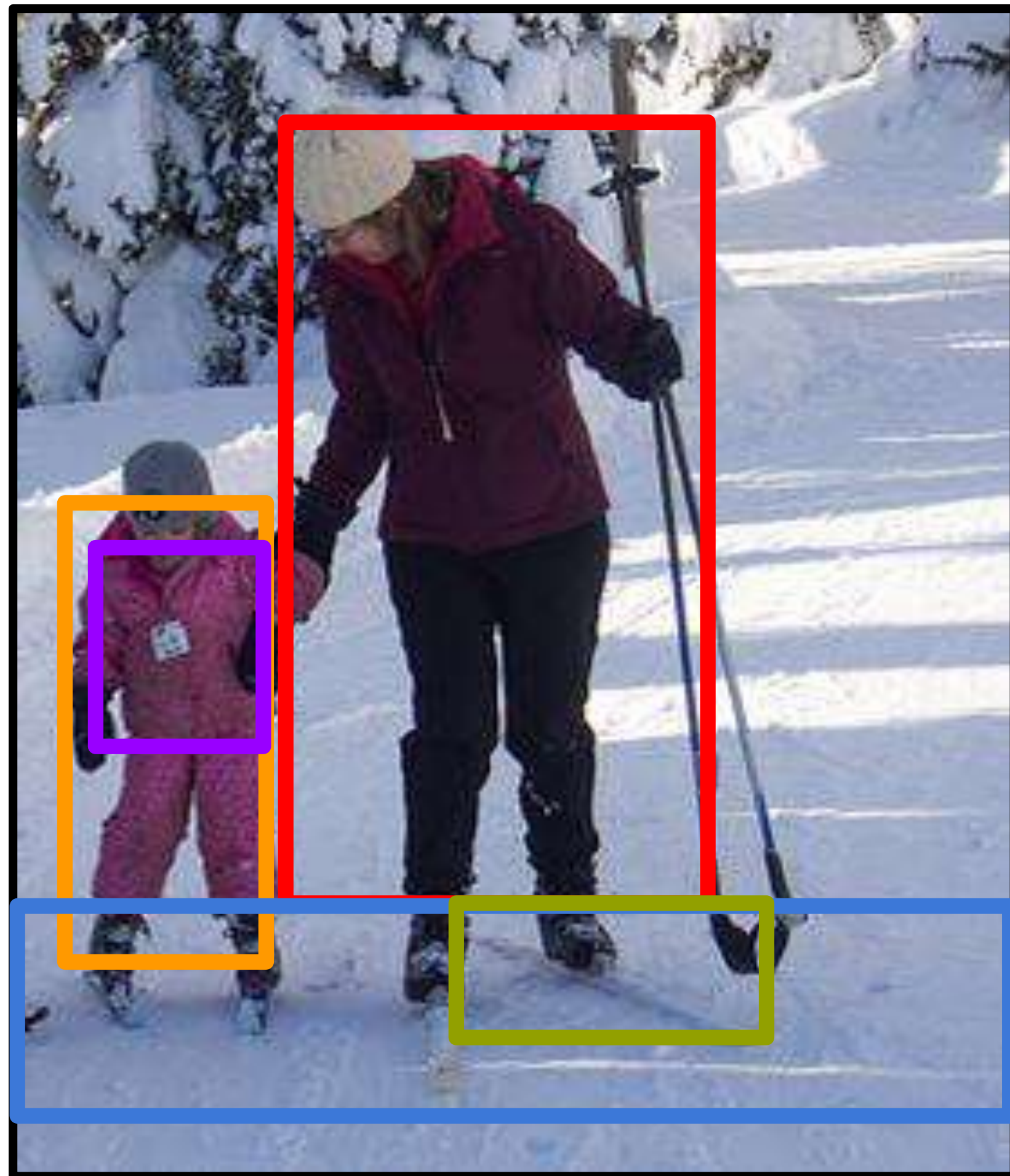
2012 to Present: Deep Learning is Everywhere

Video Understanding and Activity Recognition



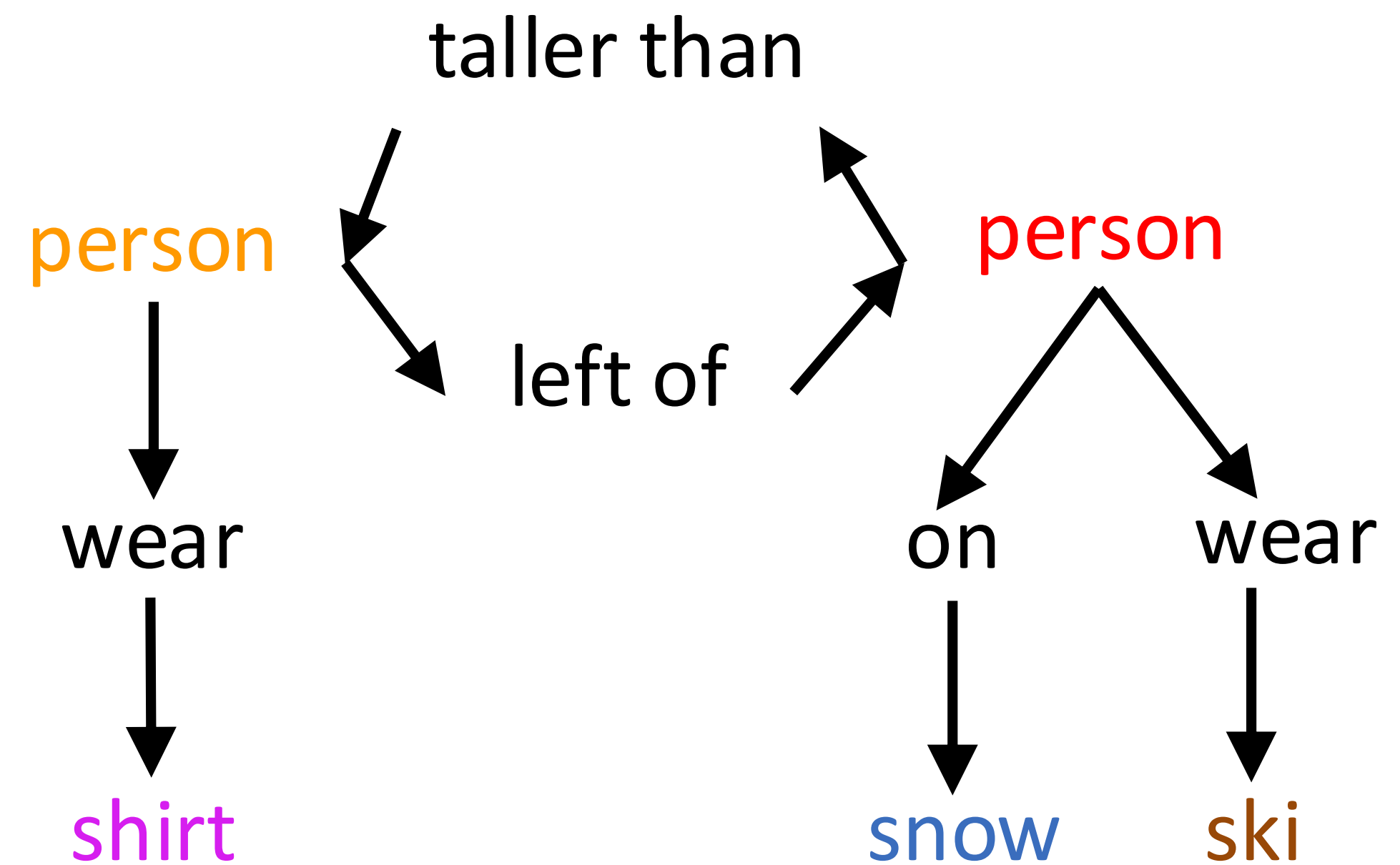
A. Gupta, J. W. J. Deng & L. Fei-Fei, *NeurIPS*, 2023, Z. Luo, Z. Durante, L. Li, W. Xie, R. Liu, E. Jin, Z. Huang, L. Y. Li, J. Wu, J. C. Niebles, E. Adeli, & L. Fei-Fei, *NeurIPS*, 2022

2012 to Present: Deep Learning is Everywhere



Results:

spatial, comparative, asymmetrical, verb,
prepositional



Krishna*, Lu*, Bernstein, Fei-Fei, *ECCV* 2016

from recognizing isolated objects toward understanding the structure and semantics of an entire scene

2012 to Present: Deep Learning is Everywhere

Early Image Captioning

Vinyals et al, 2015

Karpathy and Fei-Fei, 2015



*A white teddy bear
sitting in the grass*



*A man in a baseball
uniform throwing a ball*



*A woman is holding
a cat in her hand*



*A man riding a wave
on top of a surfboard*



*A cat sitting on a
suitcase on the floor*



*A woman standing on a
beach holding a surfboard*

All images are CC0 Public domain:
<http://pixabay.com/en/luggage-antique-cat-1643010/>
<http://pixabay.com/en/teddy-plush-bear-cute-teddy-bear-1623436/>
<http://pixabay.com/en/surf-wave-summer-sport-lit-1668716/>
<http://pixabay.com/en/woman-female-model-portrait-adult-983967/>
<http://pixabay.com/en/handstand-lake-meditation-496008/>
<http://pixabay.com/en/baseball-player-shortstop-in-field-1045263/>

Captions generated by Justin Johnson using [NeuralTalk2](#).

to connect vision with language

2012 to Present: Deep Learning is Everywhere

Modern Image Captioning – can be much more detailed!



A white teddy bear sitting in the grass

Gemini 3 (2026):

The image features a charming, **cream-colored teddy bear** sitting peacefully in a sunlit grassy field.

Key Details:

The Bear: It has soft, fluffy fur and is wearing a **hand-knitted white sweater** or dress. Its expression is neutral yet sweet, with dark eyes and a stitched brown nose.

The Setting: The bear is positioned in lush green grass dotted with small white **daisies**. One particular daisy appears to be "growing" or resting right on top of the bear's head, giving it a whimsical look.

Lighting: The scene is bright and warm, suggesting a clear, sunny day. The sunlight creates soft shadows and highlights the texture of both the bear's fur and the knitted yarn of its outfit.

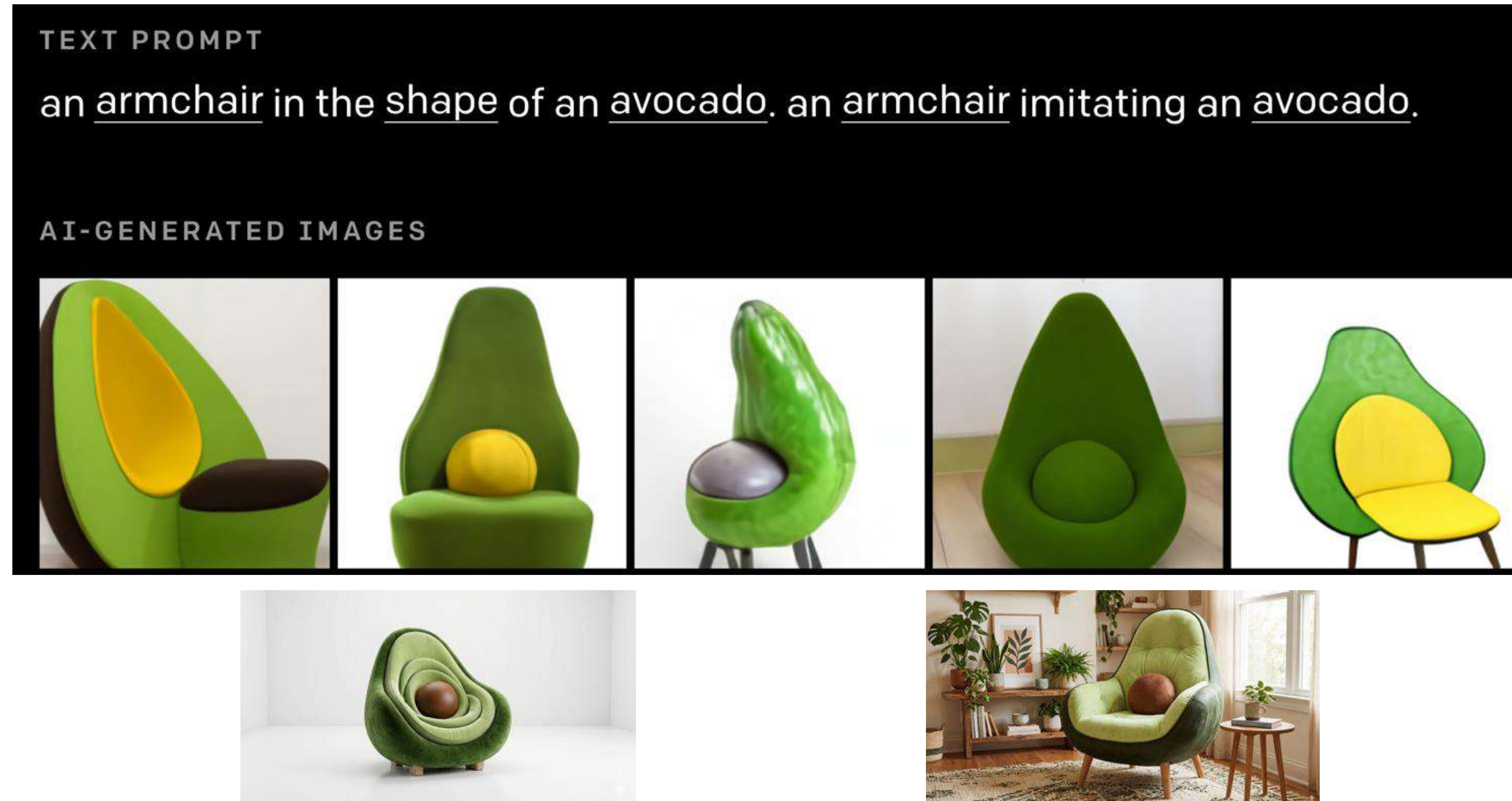
The overall mood is serene, nostalgic, and gentle.

All images are CC0 Public domain:
<https://pixabay.com/en/luggage-antique-cat-1643010/>
<https://pixabay.com/en/teddy-plush-bear-cute-teddy-bear-1623436/>
<https://pixabay.com/en/surf-wave-summer-sport-litoral-1668716/>
<https://pixabay.com/en/woman-female-model-portrait-adult-983967/>
<https://pixabay.com/en/handstand-lake-meditation-496008/>
<https://pixabay.com/en/baseball-player-shortstop-in-field-1045263/>

Captions generated by Justin Johnson using NeuralTalk?

describe objects, attributes, relationships, lighting, context

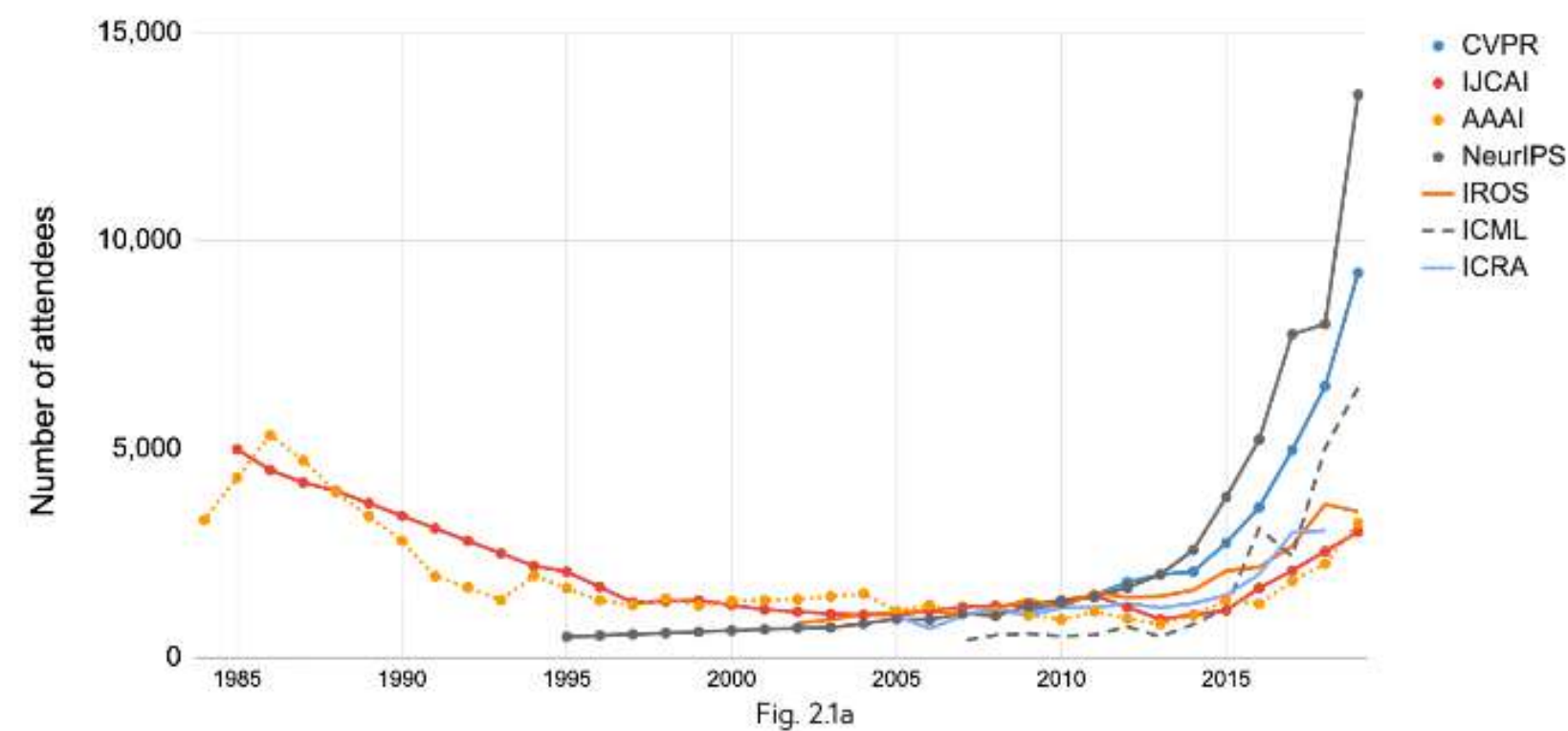
2012 to Present: Deep Learning is Everywhere



Ramesh et al, "DALL·E: Creating Images from Text", 2021. <https://openai.com/blog/dall-e/>
Remaining images generated by GPT-4o and Imagen 4

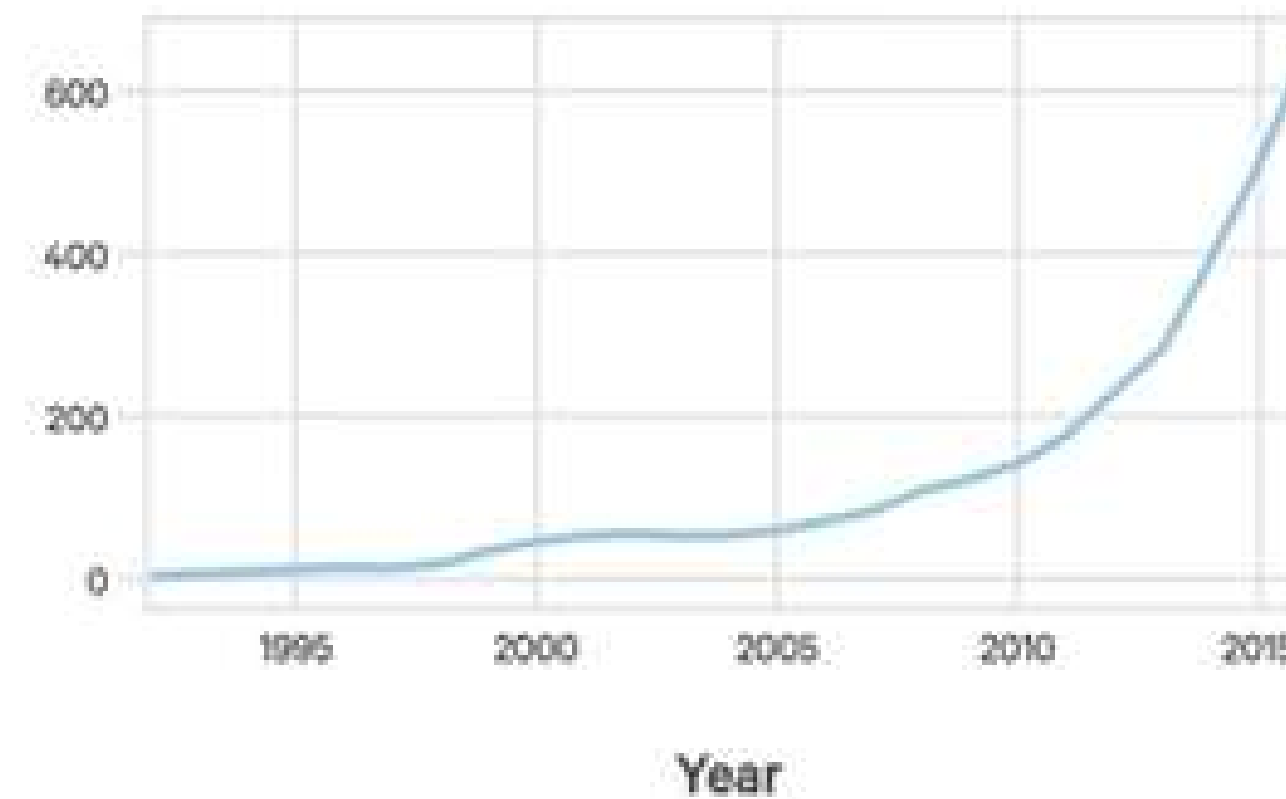
AI's Explosive Growth & Impact

Attendance at large conferences (1984-2019)
Source: Conference provided data.



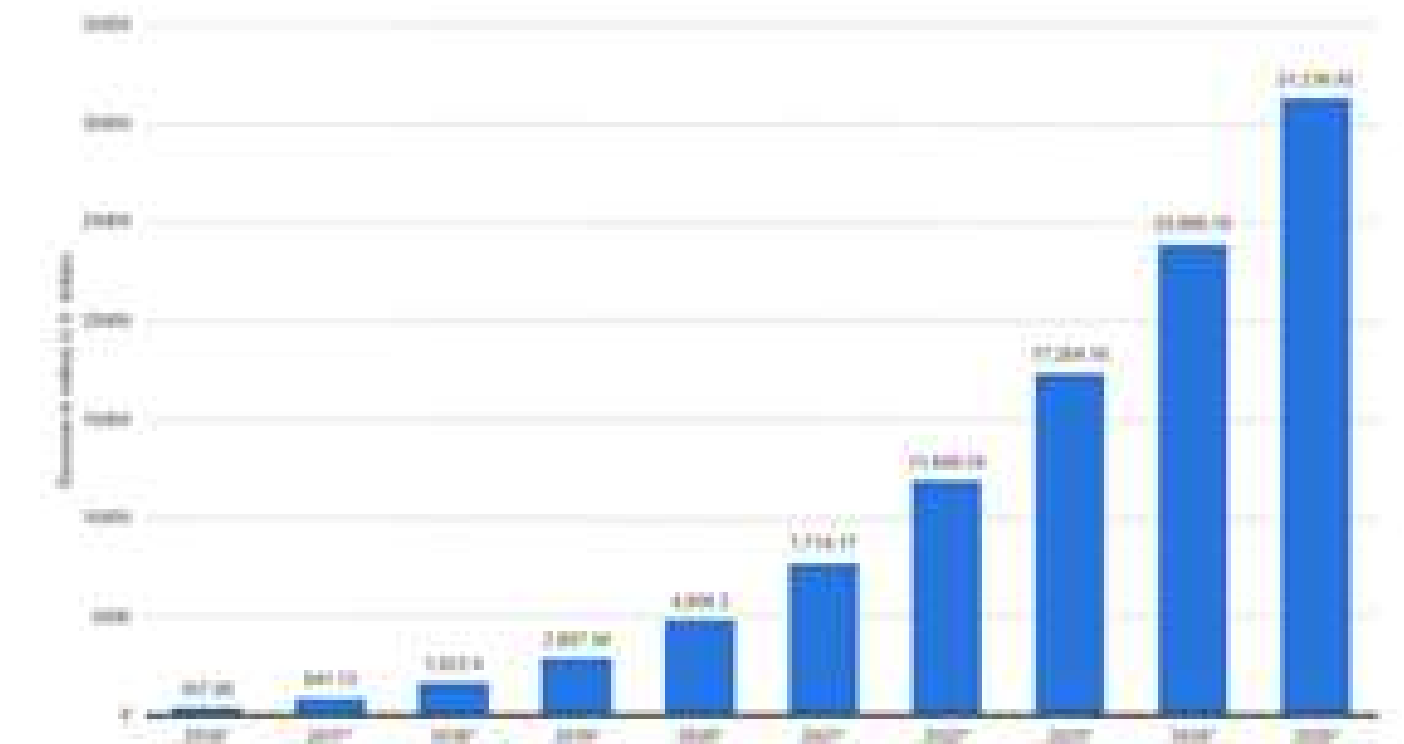
**Number of attendance
At AI conferences**

Source: The Gradient



**Startups Developing AI
Systems**

Source: Crunchbase, VentureSource, Sand
Hill Econometrics



**Enterprise Application AI
Revenue**

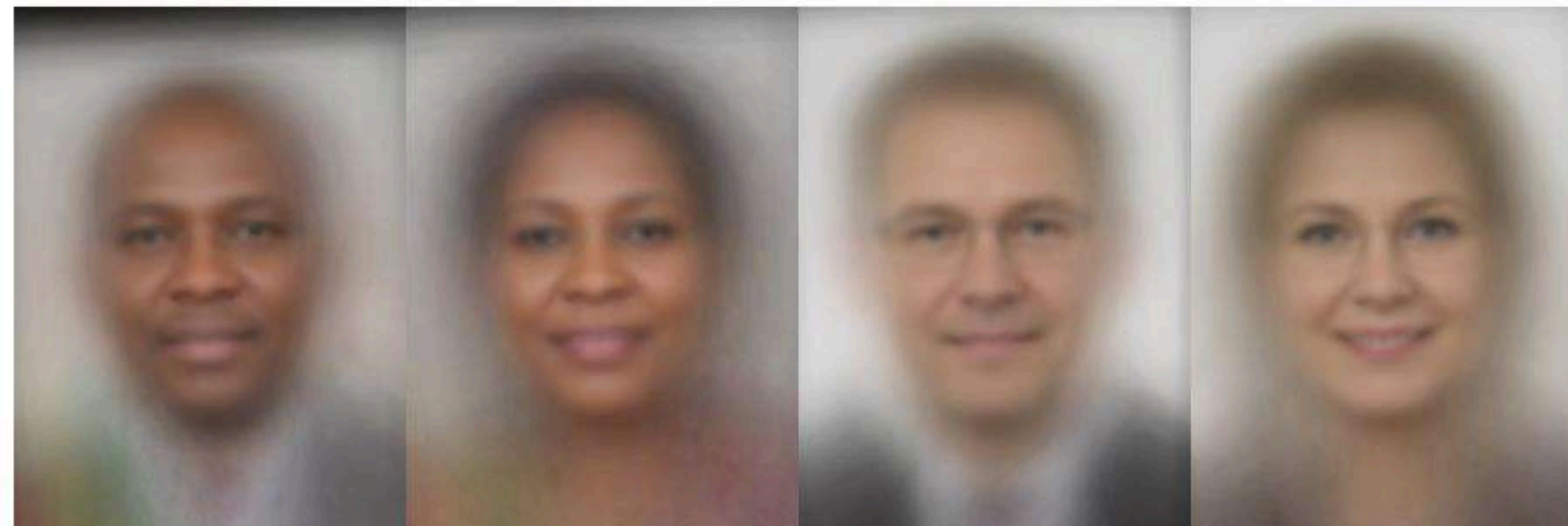
Source: Statista

Despite the successes, computer
vision still has a long way to go

Computer Vision Can Cause Harm

Harmful Stereotypes

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
 Microsoft	94.0% 	79.2% 	100% 	98.3% 	20.8%
 FACE++	99.3% 	65.5% 	99.2% 	94.0% 	33.8%
 IBM	88.0% 	65.3% 	99.7% 	92.9% 	34.4%

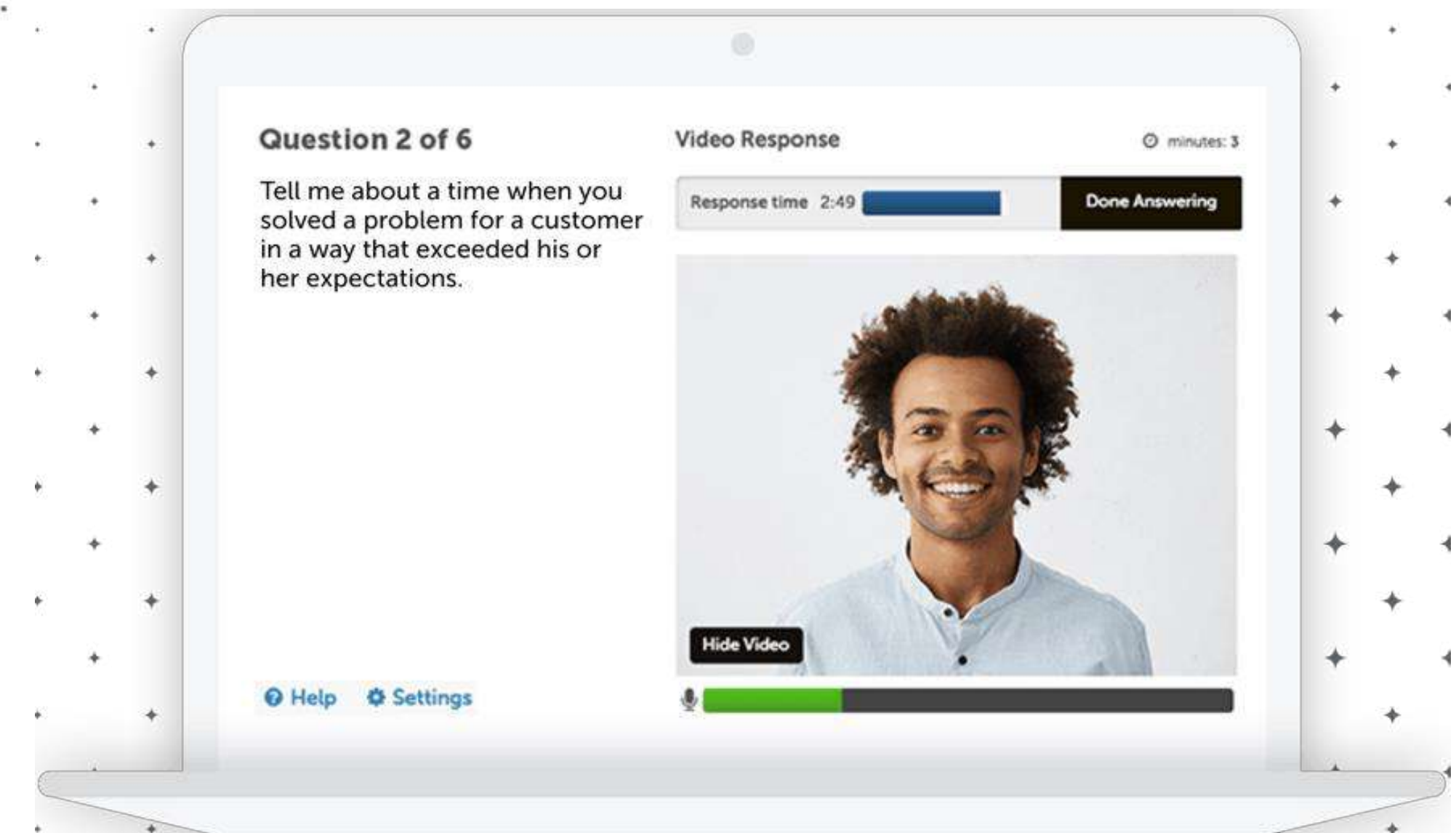


Affect people's lives

Technology

A face-scanning algorithm increasingly decides whether you deserve the job

HireVue claims it uses artificial intelligence to decide who's best for a job. Outside experts call it 'profoundly disturbing.'



Source: <https://www.washingtonpost.com/technology/2019/10/22/ai-hiring-face-scanning-algorithm-increasingly-decides-whether-you-deserve-job/>
<https://www.hirevue.com/platform/online-video-interviewing-software>

Computer Vision Can Save Lives



Computer Vision Can Save Lives



R. Zhang, S. Lee, M. Hwang, A. Hiranaka, C. Wang, W. Ai, J. J. Tan, S. Gupta, Y. Hao, G. Levine, R. Gao, A. Norcia, L. Fei-Fei, J. Wu, *CoRL*, 2023

And there is a lot we don't know how to do

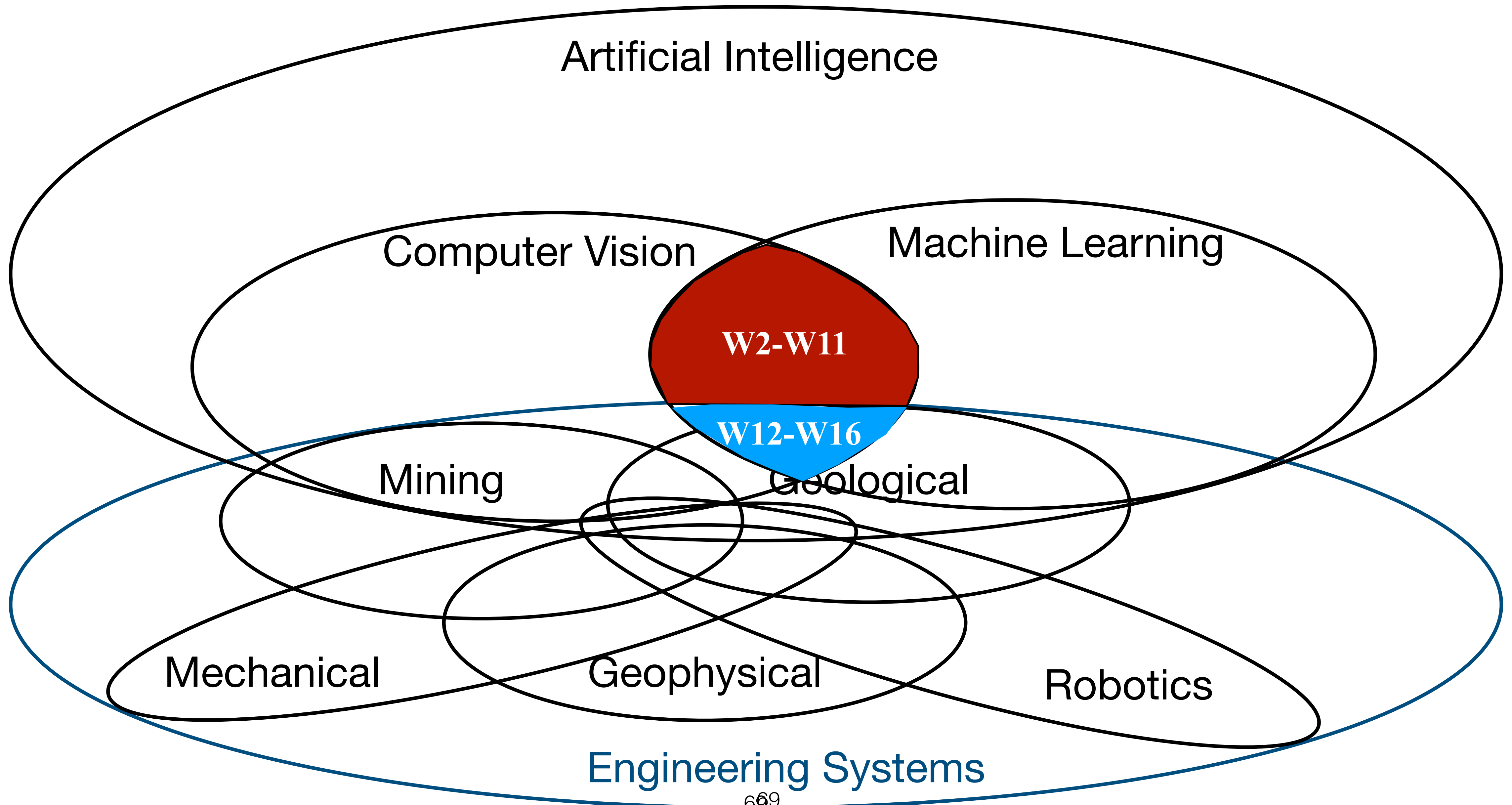


https://fedandfit.com/wp-content/uploads/2020/06/summer-activities-for-kids_optimized-scaled.jpeg



This image is
copyright-free United
States government
work

Next Lecture: CV for Engineering



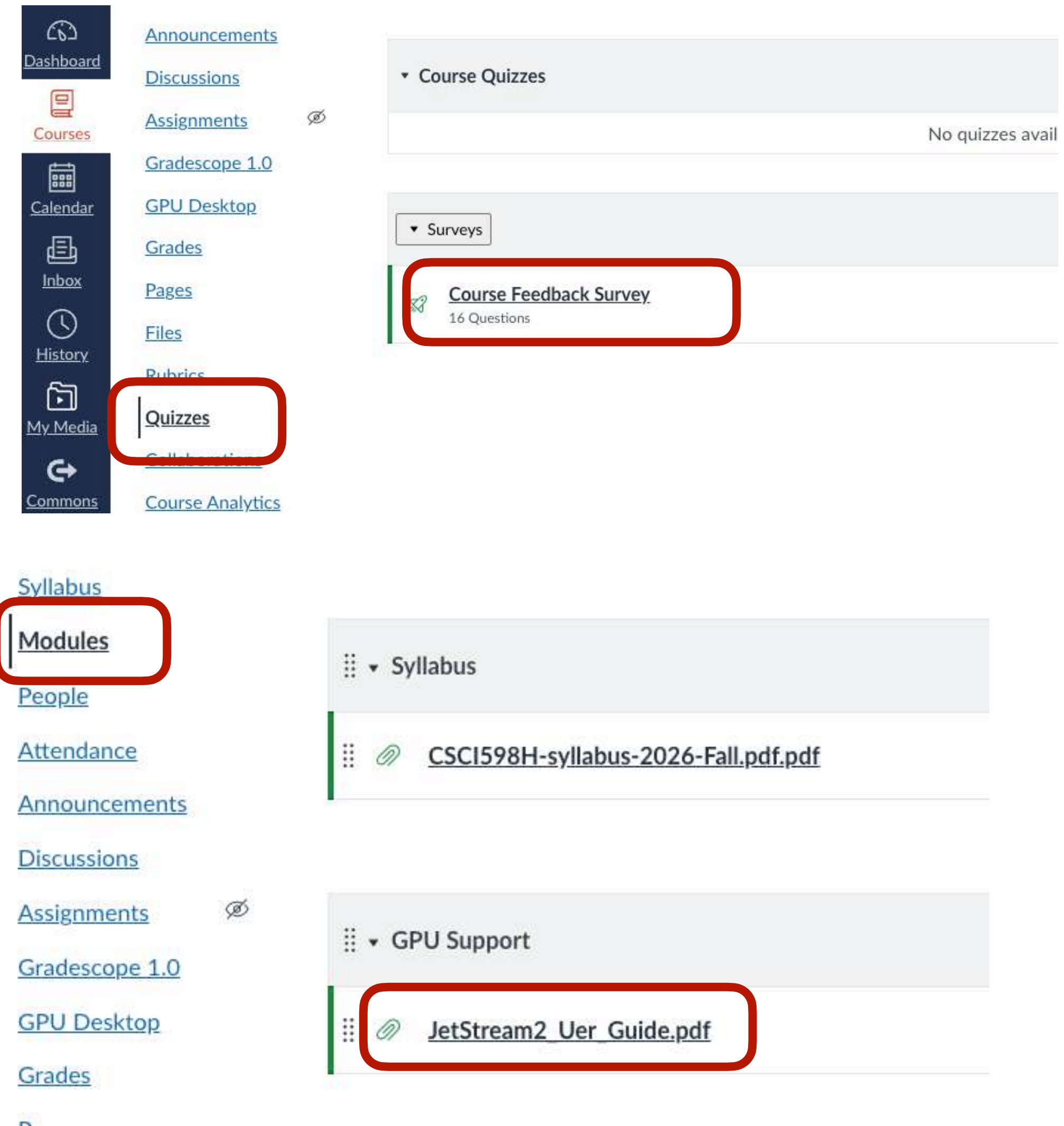
Course Feedback & Homework

- Course feedback on Canvas:
Quizzes → Course Feedback Survey

- Homework: Setup JetStream2.
Modules → GPU Support

Official User Interfaces:

<https://docs.jetstream-cloud.org/ui/horizon/launch/#alternative-volume-backed-instance>



JetStream2 Platform

```
(base) ubuntu@mailab:~$ nvidia-smi
Tue Aug 25 17:17:35 2026

+-----+
| NVIDIA-SMI 535.216.01                  Driver Version: 535.216.01   CUDA Version: 12.2   |
+-----+-----+-----+-----+
| GPU  Name                Persistence-M | Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |      Memory-Usage | GPU-Util  Compute M. |
|                                           | MIG M.         |
+-----+-----+-----+-----+
|   0  GRID A100X-20C        On          | 00000000:04:00.0 Off |          0          |
| N/A   N/A   P0              N/A /  N/A |   0MiB / 20480MiB |    0%      Default  |
|                                           | Disabled       |
+-----+-----+-----+-----+

+-----+
| Processes: |
| GPU  GI   CI        PID   Type   Process name                      GPU Memory |
|      ID   ID                                   |             Usage |
+-----+-----+-----+-----+
| No running processes found |
+-----+

(base) ubuntu@mailab:~$
```

JetStream2 Platform

API Access

Compute

Overview

Instances

Images

Key Pairs

Server Groups

Volumes

Container Infra

Network

DNS

Object Store

Share

Identity

Instances

Instance ID = Filter Launch Instance Delete Instances More Actions

Displaying 3 items

☐	Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
☐	SUSMIT	Featured-Ubuntu24-gpu-driver-fix-1738004289643138158	10.1.24.116, 149.165.168.134	g3.large	jetstream2_cs_course	Shutoff	nova	None	Shut Down	1 month	<button>Start Instance</button>
☐	pkrisna	Featured-Ubuntu24-gpu-driver-fix-1738004289643138158	10.1.24.203, 149.165.171.27	g3.large	jetstream2_cs_course	Shutoff	nova	None	Shut Down	1 month, 1 week	<button>Start Instance</button>
☐	MAILab	Featured-Ubuntu22-gpu-driver-fix-1738004259901491136	10.1.24.212, 149.165.150.227	g3.large	jetstream2_cs_course	Active	nova	None	Running	1 month, 1 week	<button>Create Snapshot</button>

Displaying 3 items

Extra Opportunities

**YOUR
BIG
YEAR®**

The following pages contain three program flyers to share on your campus:

- [AI Accelerator with IBM SkillsBuild](#)
- [IBM Z Student Ambassador Program](#)

Questions?

Contact our full team at social@yourbigyear.com
or via Text/WhatsApp: +1 201-970-2243

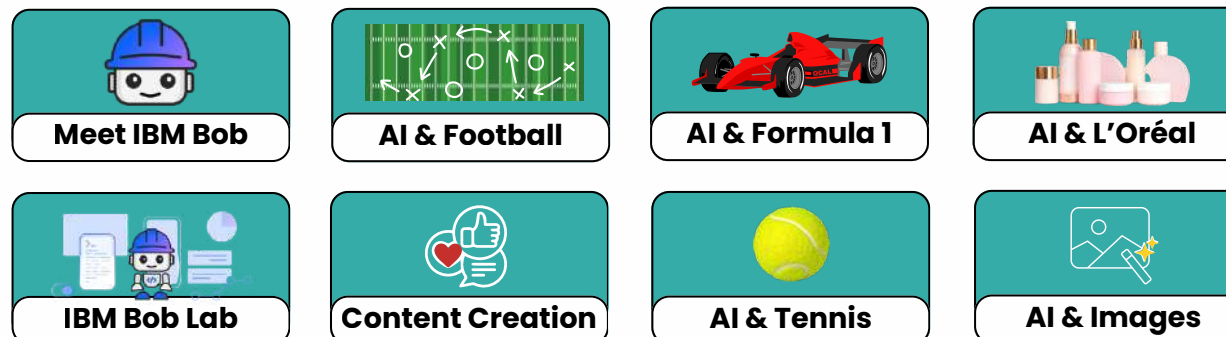
www.yourbigyear.com/university-opportunities

AI Accelerator IBM SkillsBuild + Rewards!

Build your future with the AI Accelerator. In just minutes, explore how AI is transforming sports, media, beauty, and more through hands-on micro-credentials. Earn industry-recognized badges, attend exclusive IBM Tech Talks, and gain skills that stand out on your resume and LinkedIn.

Score your favorite NFL player's Jersey, win a L'Oréal skincare bundle, get a free graphic design masterclass, and more! (T&Cs apply) Join the Accelerator!

 **Microcredentials** - complete all 8 (10-15mins each)



 **IBM Tech Talk Events** - attend at least 1 session
IBM Speakers, IBM Talent Acquisition, Tech Learning Support



 **Earn industry-recognized badge & certificate**
Submit proof of completion and receive shareable materials for LinkedIn



Complete the AI Accelerator at
yourbigyear.com/ai-campus-connect

Submission deadlines every month

University email required
No cost to participate
Terms and Conditions apply



IBM Z STUDENT AMBASSADOR PROGRAM



 **WHAT IS THE IBM Z STUDENT AMBASSADOR PROGRAM?**

IBM Z Student Ambassadors are mainframe advocates on their campus, promoting awareness of mainframe technology and careers.

They do this by establishing or partnering with clubs, hosting events, attending professional conferences, and joining a global community of professionals and students.



Check out this video
highlighting the impact of the
2025/26 Ambassador Cohort

Join an Exclusive Global Tech Community!



245+
Z Ambassadors



52+ Campus
Clubs



28+
Countries

Exclusive opportunities for ambassadors

- **Leadership & Skills:** Virtual sessions, LinkedIn guides, and industry-recognized digital credentials
- **Event Support:** Club Kits, IBM guest speakers, and social media templates
- **Exclusive Networking:** Invites to conferences, Executive Roundtables, and annual Ambassador Showcase
- **Mentorship & Community:** Direct access to IBM leaders, global peers, and personalized coaching
- **Public Recognition:** Weekly shoutouts, LinkedIn features, and talent page



HOW PROFESSORS CAN HELP

- **Refer Students:** Share this flyer and recommend applicants
- **Host Presentations:** Invite ambassadors to speak in your classes
- **Support Events:** Assist with event spaces, club budgets, and logistics

Contact: ibmz@yourbigyear.com with any questions



**SCAN OR CLICK HERE FOR APPLICATION
INFORMATION FOR STUDENTS**